

GENERAL SPECIFICATION G6.12

Instrumentation

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G6.12 INSTRUMENTATION

G6.12.1 Common Requirements

(a) General

- (i) All instruments, gauges and control equipment which perform similar duties shall be of uniform type and manufacture throughout the Works to facilitate maintenance and the economical stocking of spare parts.
- (ii) Instruments shall have a manufacturer supplied engraved/stamped 316L tag(s) bearing the instrument tag number and calibration data.
- (iii) Instruments shall be supplied complete with all brackets, stands, supporting steelwork and enclosures (separate from the instrument cases) necessary for securing them in their working positions and affording complete protection at all times including periods of servicing, adjustment, calibration and maintenance.
- (iv) Indicators and gauges mounted directly on equipment shall be sized to give full legibility when viewed from a position with convenient and easy access or from the point at which any operation requiring observation of the gauge is performed.
- (v) All instruments offered shall have a provable extensive track-record in the intended application.
- (vi) Refer to General Specification G6.2 Installation Requirements for instrument installation requirements.

(b) Environmental Resistance

- (i) Instruments shall have weatherproof and dust-proof cases. Instrument cases shall be of corrosion-resistant material or finish. The enclosures of all electronic instruments shall be rated IP 65 as a minimum unless otherwise specified.
- (ii) Moving parts and contacts shall be adequately protected from the ingress of dust.
- (iii) Unless otherwise specified instruments shall be finished in the manufacturer's standard colour.
- (iv) Dials and bezels and internal components shall be of stainless steel, bronze or other corrosion-resistant material such that no peeling or discoloration shall take place with age.
- (v) All instruments mounted outdoors shall have 316 stainless steel weather canopies (sun shades) of approved design, with adequate clearance for all maintenance and calibration activities without removal.
- (vi) Wetted parts shall be selected to achieve excellent compatibility of the material with the process fluid and service conditions in which the instrument is proposed to be used.

- (c) Signal Transmission
 - (i) Unless otherwise specified all functions shall be transmitted electrically and all analogue signal transmission systems shall be in accordance with IEC 60381-1, being isolated 4 to 20 mA dc plus HART.
 - (ii) Fieldbus connections to instruments and actuators shall not be used.
 - (iii) Where analytical instruments without HART have a fault contact this shall be connected to the SCADA system via the appropriate PLC.
- (d) Integral Indicators
 - (i) Each measuring transmitter shall have an integral LCD indicator, and shall be mounted as near to the tapping or measurement point as possible subject to the overriding considerations of environmental compatibility and safe accessibility for maintenance and servicing.
- (e) Power Supply
 - (i) Instruments shall preferably be powered by 24 Vdc or be loop-powered.
 - (ii) Field instruments requiring a 230 Vac power supply shall be permanently cabled to a UPS distribution board where there shall be a MCB dedicated to each such instrument. General purpose socket outlets shall not be used.
- (f) Analytical Instruments
 - (i) Analysers shall be logically grouped and installed together on common analyser panels. Analyser panels shall include a sample bench and other facilities for operating portable test meters.
 - (ii) Each installation shall incorporate both isolation and flow adjustment needle valves for each analyser, and a rotameter and all necessary pipework and hardware.
 - (iii) Sample transport times shall be minimised by provision of a by-pass with separate flow adjustment.
 - (iv) A tee-off sample outlet at any analyser panel, as well as tundish and drain, to allow manual sampling.
 - (v) Analyser waste shall be properly plumbed to drains and shall not simply run onto the floor.
 - (vi) If the analyser panel is more than 10 metres from the sample off-take points, manual sample facilities shall be provided at both points.
 - (vii) Analysers shall be provided complete with all reagents, chemicals, gases and all consumables and wearing parts required for continuous operation through until Substantial Completion.

G6.12.2 Signal Conditioning Devices

- (a) Signal conditioning devices shall include isolators, amplifiers, current-to-pulse and pulse-to-current converters, arithmetic modules, trip amplifiers and similar units.

- (b) Each device shall be designed for mounting within an enclosure using a mounting plate, rack or rail. An indication of "power supply on" shall be provided at the front of each unit.
- (c) Each analogue output circuit shall have independent zero and span adjustments.
- (d) Set-points on the instrument front shall be adjustable.
- (e) The states of output relays shall be indicated at the front of the unit.
- (f) Where used on main plant PLC analogue input loops, trip amplifiers shall pass through the superimposed HART signals.

G6.12.3 Pressure Transmitters and Differential Pressure Transmitters

- (a) Where necessary, a diaphragm seal shall be used to isolate the gauge tube from corrosive fluid media. In ammonia applications, the diaphragm shall be in stainless steel. In chlorine applications, the diaphragm shall be in silver or tantalum. In sulphur dioxide applications, the diaphragm shall be in tantalum.
- (b) The zero and span of a pressure transmitter shall not change by more than $\pm 0.05\%$ of the span per $^{\circ}\text{C}$ change in ambient temperature. After application for 10 minutes of pressure at 130% of maximum pressure, the change in zero and span shall not exceed $\pm 0.1\%$ of the span.
- (c) Pressure transmitters shall have an accuracy of $\pm 0.15\%$ of span, or better and shall be protected to IP 65 or higher. For transmitters installed in locations prone to flooding they shall be to IP 68 for a maximum submergence of 10 metres for 8 hours.
- (d) Pressure transmitters shall provide a 4 to 20mA HART output proportional to the pressure.
- (e) Pressure transmitters shall be able to sustain a 200% overpressure without damage.
- (f) All pressure transmitters used for pressure and level measurement shall be complete with a 2-valve isolation manifold. Vent ports shall be tubed such that the discharge does not create a hazard.
- (g) All differential pressure transmitters used for pressure and level measurement shall be complete with a 3-valve isolation manifold. Vent ports shall be tubed such that the discharge does not create a hazard.
- (h) Flange-mounted pressure/differential pressure transmitters used for pressure and level measurement shall be complete with an isolation valve and flushing facilities.

G6.12.4 Differential Pressure Transmitters Used for Flow Measurement

- (a) Flow transducers of the differential pressure type shall be designed and installed in compliance with ISO 7145. Primary devices shall be either carrier-ring type orifice assemblies with a stainless-steel orifice plate or pitot / venturi tubes. All materials shall be appropriate to the metered fluid and service conditions.

- (b) Where necessary, a special diaphragm shall be used to segregate the sensor from corrosive fluid media. In ammonia applications, the diaphragm shall be in stainless steel. In chlorine applications, the diaphragm shall be in silver or tantalum. In sulphur dioxide applications, the diaphragm shall be in tantalum.
- (c) Orifice assemblies shall have identification tags showing the direction of flow, orifice diameter and position of drain hole.
- (d) Differential pressure transmitters shall have over-range protection up to 1.5 times the maximum line pressure.
- (e) They shall be adjacent to the primary element.
- (f) Differential pressure transmitters used for flow measurement shall be complete with a 3-valve isolation manifold. Vent ports shall be tubed such that the discharge does not create a hazard.
- (g) Where applicable, an integral square root extraction device shall provide a 4-20 mA HART output signal linearly proportional to flow.
- (h) Differential pressure transmitters shall have a pressure measurement accuracy of $\pm 0.15\%$ of span or better. Flow measurement accuracy shall be better than $\pm 2\%$ of span.
- (i) For steam flow measurement, multivariable transmitters shall be used such that transmitter output is normalised steam mass flow.

G6.12.5 Pressure Switches and Differential Pressure Switches

- (a) The pressure values at which pressure and differential pressure switches operate shall be fully adjustable over the whole range of the instrument and the set value shall be clearly indicated by means of a scale and pointer or be set electronically. Pressure and differential pressure switches shall have over range protection. Set point repeatability shall be $\pm 1\%$ of adjustable range or better.
- (b) Pressure switches and differential pressure switches for pressures up to 1bar should preferably be piezoelectric type. They shall be installed minimizing any kind of mechanical disturbance produced by the process, e.g. vibration, water hammer, etc.
- (c) Where necessary, a diaphragm seal shall be used to segregate the switch tube from corrosive fluid media. In ammonia applications, the diaphragm shall be in stainless steel. In chlorine applications, the diaphragm shall be in silver or tantalum. In sulphur dioxide applications, the diaphragm shall be in tantalum.
- (d) Pressure and differential pressure switches shall be housed in cast bronze or aluminium alloy enclosures. The degree of protection shall be IP 65 except for locations prone to flooding where it shall be IP 68 with a maximum submergence of 10 metres for 8 hours.
- (e) Pressure and differential pressure switches on chemical service shall be provided with an external block and bleed isolation arrangement, and the vent from the bleed valve shall be directed safety.
- (f) Where pressure switches and differential pressure switches are specified as being "Electronic Pressure Switches" they shall be four-wire devices operating from 24 Vdc power, and having relay outputs with selectable failure states.

- (g) Electronic pressure switches shall have trip points which are adjustable to a specific pressure value displayed on an LCD display.

G6.12.6 Sight Glass Level Indicators

- (a) Sight glass level indicators shall be provided with a protective housing, borosilicate glass tube and isolating valves.
- (b) The gauge shall be suitably calibrated in volumetric SI. units unless otherwise specified that level units are required.
- (c) Where necessary, the gauge shall be provided with flushing and drain points with valves and hose connections.

G6.12.7 Hydrostatic Level Transmitters (Externally mounted)

- (a) Hydro-static level transmitters shall be flange-mounted on the vessel and shall comprise a diaphragm operated sensor and an electronics module. The materials of construction shall be compatible with the fluid with which they are in contact.
- (b) Where necessary, a special diaphragm shall be used to segregate the sensor from corrosive fluid media.
- (c) The output of the transmitter shall be 4-20 mA HART proportional to liquid level. Where the transmitter is fitted to a horizontal cylindrical storage tank for contents measurement, a converter shall be incorporated in the transmitter itself or in the signal loop to give a linear output proportional to contents.
- (d) The electronics module with all adjustments and terminations shall be enclosed in a housing with degree of protection of IP 65. For locations prone to flooding, degree of protection shall be IP 68 with a minimum submergence of 10 metres for 8 hours.
- (e) The transmitter accuracy shall be $\pm 0.25\%$ fsd or better. The zero and span of a pressure transmitter shall not change by more than $\pm 0.05\%$ of the span per $^{\circ}\text{C}$ change in ambient temperature.
- (f) Hydro-static level transmitters used for sludge or chemical applications shall be complete with an isolation valve, drain valve, and flushing facilities.

G6.12.8 Hydrostatic Level Transmitters (Immersed)

- (a) The hydrostatic level transmitter shall comprise an immersion type level sensor suspended on a cable, with a signal processor in a weatherproof enclosure. The diaphragm of the sensor shall be installed at the specified level below the zero level for the measured span. The sensor shall have a chain for retrieving it, and a weight for holding the sensor in place in the well.
- (b) The sensor wetted part shall be constructed from stainless steel. The cable shall have a protective sheath, and shall include a breather tube for the sensor. The tube shall be arranged to prevent the entry of water or condensate into the tube.

- (c) The sensor shall generate a 4-20mA HART signal linear with the measured level. The accuracy shall be $\pm 0.5\%$ of full set range, including the effects of linearity and repeatability.

G6.12.9 Float Operated Level Indicators

- (a) Float operated level indicators shall comprise a float operating on fixed guide wires. The float shall be connected by a stainless-steel cable to an indicator unit which moves up and down a marker board fitted to the external wall of the tank as the product level falls and rises. Pulleys shall be of stainless steel construction.
- (b) Switch assemblies actuated by the moving indicator may be fitted to the marker board to provide high or low alarms and control initiation.
- (c) Indicator scale shall be of aluminium, brass or cedar wood construction graduated in metric linear or volumetric units.
- (d) The range of the instrument shall be 0 up to 20m (or equivalent volumetric units) and the accuracy/repeatability shall be $\pm 10\text{mm}$.

G6.12.10 Float Gauging Level Transmitters

- (a) Float gauging level transmitters shall comprise a float and tension device. Floats shall be of stainless steel or as approved by S.O..
- (b) The float shall be suspended by a stainless-steel wire or perforated tape which shall operate the transmitter installed above the float stilling well or measuring chamber.
- (c) The transmitter shall provide a 4 to 20mA HART output signal proportional to the level measurement range and shall be housed in an enclosure protected to IP 65. For locations prone to flooding the enclosure shall be protected to IP 68 with a minimum submergence of 10 m for 8 hours.
- (d) The accuracy shall be $\pm 0.35\%$ or better.

G6.12.11 Ultrasonic Level Transmitters (including for flow measurement)

- (a) Ultrasonic level transmitters shall comprise a sensor and control unit though these may be combined providing the safe access requirements specified in General Specifications G6.2 Instrument Installation are met.
- (b) The sensor shall be suitable for flange or bracket mounting as required and have a minimum degree of protection of IP 67. The sensor shall incorporate a temperature sensing element.
- (c) The accuracy of the instrument shall be $\pm 0.25\%$ or better of range for ranges of 2400 mm or more. For ranges less than 2400 mm, the accuracy shall be $\pm 6\text{ mm}$ or better. The instrument shall incorporate automatic temperature compensation.

- (d) The associated transmitter shall be mounted in an enclosure with protection to IP 65. For locations prone to flooding the enclosure shall be protected to IP68 with a minimum submergence of 10 m for 8 hours. It shall be programmable with an integral programming keyboard, digital display, relays for alarm, control and system fault and shall provide an isolated 4 to 20mA HART output signal. The transmitter shall contain signal processing facilities to eliminate the effects of standing echoes from tank features, moving echoes from mixers and other such problems.
- (e) For flow measurement applications, a totalizer pulse output shall be provided.
- (f) The design and application of ultrasonic level meters shall consider the vessel or channel construction, the material, size, shape, environment, process fluid or material, the presence of foam, granules, size etc.
- (g) The installation shall avoid any degradation of performance from spurious reflections, absorption, sound velocity variations, sensor detection area, temperature fluctuations, specific gravity changes and condensation.
- (h) If turbulence exists, shielding, stilling tubes or other measures shall be provided to avoid effects on the measurement.

G6.12.12 Conductive Probe Level Switches

- (a) The level switch system shall consist of one or more sensing electrodes and controller containing switching circuitry and control relay(s).
- (b) Electrode rods shall be stainless steel, or other material suitable for application. For applications exceeding 2m length, insulated wire rope type probes may be used for applications with no turbulence. Materials used for electrode heads shall be corrosion proof.
- (c) Controllers located within panels shall be of the chassis type and shall have a minimum degree of protection of IP20, adjustable sensitivity and response time settings, facility for setting fail safe operation at high or low level, light emitting diode output relay operating indication and double pole voltage-free changeover type output contacts. The controller shall be supplied from the mains.
- (d) Controllers not mounted in panels shall be provided with surface mounting enclosures having a degree of protection of IP 65 except for locations prone to flooding the enclosure shall be protected to IP 68 with a minimum submergence of 10 metres for 8 hours.
- (e) Indicators for power and switch status shall be provided and shall be visible with all covers closed.
- (f) Level switches shall be 24V dc powered. Outputs may be PNP or relay contact.
- (g) State of the output upon internally detected failure shall be configurable.

G6.12.13 Radar Level Transmitters

- (a) Radar level transmitters shall comprise a sensor and control unit though these may be combined providing the safe access requirements specified in General Specification G6.2 Installation Requirements are still met.
- (b) The sensor shall be suitable for flange or bracket mounting as required, have an antenna of PTFE or Alloy C-22, of a condensation-shedding design, and have a minimum degree of protection of IP 67.
- (c) Radar level transmitters shall operate either at:
 - (i) 26 GHz with an accuracy of $\pm 2\text{mm}$ or better; or
 - (ii) 80 GHz with an accuracy of $\pm 1\text{mm}$ or better.
- (d) The associated transmitter shall be mounted in an enclosure with protection to IP 65. It shall be programmable with an integral programming keyboard, have a digital display, and shall provide an isolated 4 to 20mA HART output signal. The transmitter shall contain signal processing facilities so as to eliminate the effects of standing echoes from tank features, moving echoes from mixers and other such problems.
- (e) The design and application of radar level meters shall take into account the vessel or channel construction, the material, size, shape, environment, process fluid or material, the presence of foam, granule size etc.
- (f) Radar level transmitters shall comply with local telecommunications and radio-frequency regulations.

G6.12.14 Tuning Fork Level Switches

- (a) Wetted materials shall be suitable for the process fluid.
- (b) The level switch shall be a one piece unit having an enclosure to IP65 or better.
- (c) The correct operation of the level switch shall be resistant to build-up on the forks and the level switch shall be able to be adjusted so as to not detect foam.
- (d) Level switches shall be 24V dc powered, 4-wire devices. Outputs may be PNP or relay contact.
- (e) State of the output upon internally detected failure shall be configurable.
- (f) State of the output shall be indicated by means of visible LED or similar.

G6.12.15 Capacitive Level Switches

- (a) The capacitive level switch sensing probe shall either screw into a boss on the side of a vessel or shall be suspended from above. When immersed by the rising process fluid, the capacitance will be altered enabling the instrument to detect the presence of the fluid. The wetted materials shall be suitable for the process fluid. The level switch shall be a one piece unit having an enclosure to IP65 or better.
- (b) The correct operation of the level switch shall be resistant to build-up on the probe.

- (c) Level switches shall be 24V dc powered, 4-wire devices. Outputs may be PNP or relay contact.
- (d) State of the output upon internally detected failure shall be configurable.
- (e) Status of the output shall be indicated by means of visible LED or similar.

G6.12.16 RF Admittance (Capacitive) Level Transmitters

- (a) RF admittance (capacitive) level transmitters shall comprise separate sensor and transmitter units. These two units may be combined providing the safe access requirements specified in General Specification G6.2 Installation Requirements are still met.
- (b) The sensor shall be suitable for flange or bracket mounting, being suspended from the flange or bracket above. When immersed in the process fluid, the capacitance will be altered enabling the instrument to detect the vertical position of the fluid surface.
- (c) The wetted materials shall be suitable for the process fluid. Accurate measurement shall be resistant to build-up on the probe.
- (d) The design and application of RF admittance level transmitters shall take into account the vessel or channel construction, the material, size, shape, environment, process fluid or material, the presence of foam, granule size etc.
- (e) The transmitter enclosure shall be rated to IP 65.
- (f) The transmitter shall have a digital display, and shall provide an isolated 4 to 20mA HART output signal.

G6.12.17 Temperature Transmitters

- (a) Thermowells & Elements
 - (i) Temperature elements shall be in sheath tubes and in thermowells of suitable material. Each temperature sensor, unless otherwise specified, shall have a stainless steel thermowell and extension assembly, corrosion proof metal sheath and waterproof terminal head. Thermowells made of Tungsten carbide shall be used for highly abrasive application. Pockets for steam, oil and pressurised water lines shall be welded; pockets for other duties shall be screwed.
 - (ii) In general for temperature measurements up to 300 deg C, Platinum resistance temperature detectors shall be used unless the area is prone to vibration. For temperature measurement above 300 deg C up to 750 deg C, Chromel-Alumel (K) type thermocouple shall be used. For high temperature applications, noble-metal thermocouple (TC) such as Pt Rh-Pt K type in Inconel sheathed thermowell shall be used.
 - (iii) The sensor assembly shall be designed to permit removal of the temperature element without twisting the leads.
 - (iv) Temperature elements shall be provided with corresponding temperature transmitters except for motor winding RTD's which may be connected directed to motor protection devices.

- (b) Temperature transmitters
 - (i) Temperature transmitters shall have an accuracy of $\pm 0.15\%$ of span, or better and shall be protected to IP 65 or higher. For transmitters installed in locations prone to flooding they shall be to IP 68 for a maximum submergence of 10 metres for 8 hours.
 - (ii) Temperature transmitters shall provide a 4 to 20mA HART output proportional to the temperature.
 - (iii) Terminal switch & amplifier enclosures shall have a degree of protection of IP65 or better.
 - (iv) Temperature transmitters shall have selectable upscale/downscale sensor burnout protection.
- (c) Platinum resistance temperature sensors (RTD's)
 - (i) RTD's shall comply with BS EN 60751 or an equivalent standard and shall have fundamental intervals of not less than 0.385 Ohms/ $^{\circ}\text{C}$. Terminal heads and amplifiers shall be designed for four-wire connections between head and amplifier.
 - (ii) RTD elements shall be fully encapsulated and the elements and lead wires shall be hermetically sealed. The associated resistance- to-current converters shall have zero and span adjustments and input-output circuit isolation.
- (d) Thermocouples
 - (i) Thermocouples shall be of the mineral insulated type and unless otherwise specified shall be of the chromel-alumel (nickel- chromium v nickel-aluminium) type and shall comply with BS 1041 and BS EN 60584-3. Wires shall not be smaller than 16 SWG. Thermocouple junctions shall be welded. Ceramic-insulation material may be used for base metal thermocouples but low-silicon insulation material shall be used for noble metal thermocouples. Thermocouple systems shall have thermoelectric ice point reference chambers or receivers or amplifiers with automatic cold junction compensation.
 - (ii) Thermocouple receivers and amplifiers shall also have zero and span adjustment, common and series mode interference rejection circuits, radio-frequency filters, input-output circuit isolation and thermocouple break detection feature whereby the output is driven to zero or full scale as specified when the receiver or amplifier input circuit is broken.
 - (iii) Thermocouple elements shall be electrically isolated from their sheaths but each terminal head shall have facilities for earthing the thermocouple and for terminating the shield of the extension or compensation cable. Amplifier chassis shall have facilities both for being earthed to the instrument case via a capacitor and for being electrically isolated from the instrument case.

G6.12.18 Temperature Switches

- (a) Temperature switches shall have operating points fully adjustable over the whole range of the instrument and the set-value shall be clearly indicated by a dial and pointer or be electronically set.
- (b) Set point repeatability shall be $\pm 1\%$ of adjustable range or better.

G6.12.19 Variable Area Flowmeters (Rotameters)

- (a) Variable area flowmeters shall have a float moving vertically in a tapered tube, the position of the float being proportional to the flow. The meter tube shall be calibrated for the specific fluid in mass or flow units. The calibration conditions shall be engraved on the tube. Linearity shall be $\pm 3\%$ of full scale or better and repeatability shall be $\pm 1\%$ of full scale or better across a 10:1 flow range.
- (b) The flowmeter shall have a glass or stainless steel tube according to the particular application. When the use of plastic materials is recommended by the manufacturer, it shall require approval by the S.O..
- (c) Metering tubes shall be removable for range change or cleaning without dismantling the meter or removing it from the line. Metering tubes shall have ends of equal cross-sectional area and if O ring seals are used, tube retainer springs shall be outside the fluid stream. End fittings shall be rotatable to any angle.
- (d) Glass metering tubes shall be of borosilicate glass and shall be adequately shielded with safety glass on the reading side and amply vented on sides, back and bottom.
- (e) Variable area flowmeters shall be installed in locations free from vibration and with sufficient clearance for the removal of the float. If used in conjunction with a flow regulating needle valve, the valve and meter shall be close together, with the valve downstream of the meter. Pipework shall be supported, having regard to the weight of the meter.
- (f) If fitted with switches, the terminal enclosures shall have a degree of protection of IP65 except for locations prone to flooding the enclosure shall be protected to IP68 with a minimum submergence of 10 metres for 8 hours.
- (g) Position of the switches shall be fully adjustable for all the length of the rotameter.

G6.12.20 Electromagnetic Flowmeters

- (a) Construction
 - (i) Unless specified otherwise in Particular Specifications P6 ICA, electromagnetic flowmeters shall be full-bore inline types.
 - (ii) Flowmeters shall operate on the electromagnetic induction principle and shall consist of a measuring sensor and measuring transmitter complying with BS EN ISO 6817. The flow metering system shall provide pulse and analogue current (4-20 mA HART) outputs proportional to volume and rate of flow respectively.

- (iii) Measuring sensors shall have a metallic metering tube and non-conductive, abrasion-resistant lining to suit the fluid being metered. Stainless steel is preferred for metering tubes, and any deviation shall require approval by the S.O.. For potable water applications, the lining material shall be of a potable water approved material. For other applications the lining material shall be suitable for the fluid being measured. Flange material shall be the same as the metering tube.
- (iv) The flowmeter shall be suitable for use in the prevailing process conditions including but not limited to vacuum conditions where these exist.
- (v) Measuring sensors shall be fitted with an anti-roll system to prevent damage during storage and shall be flanged to BS EN 1092. The flanges shall be compatible with those specified for the associated pipework.
- (vi) Measuring sensors shall have factory-sealed power and signal cables, and where applicable, earth cables. The cable lengths shall be sufficient to permit termination external to the chamber, directly to the measuring transmitter. If wiring works are performed on site, or the original wiring arrangement needs to be modified, terminal box shall be potted and sealed with an adequate resin.
- (vii) Measuring sensors installed within a chamber shall have a degree of protection IP 68 and shall be suitable for indefinite submersion under a head of water equal to the chamber depth or 3 metres whichever is the greater.
- (viii) Unless specified otherwise in Particular Specifications P6 ICA, measuring sensors shall be installed with upstream and downstream straight pipe lengths not less than those recommended by the manufacturer. Any tapered pipe reductions upstream/downstream must comply with the flowmeter manufacturers recommendations.
- (ix) When fitted in lined non-metallic or internally-coated pipework, measuring sensors shall have an earthing electrode or corrosion-resistant earthing rings.
- (x) Measuring sensors shall be bonded and earthed in accordance with the manufacturer's instructions.
- (xi) Measuring sensors installed in a cathodic protected pipeline shall have isolation and bonding in accordance with the recommendations of the manufacturer.
- (xii) The signal processing facilities of the converter shall ensure that the output signals are unaffected by interfering voltages, stratified flow, changes in fluid electrical conductivity within the limit stated, non-homogeneity of the fluid and the presence of ferrous particles. The instrument zero shall be set at the manufacturer's premises and thereafter this zero shall be maintained automatically without interruption of flow or output signals. The zero and output signals shall be unaffected by partly-fouled electrodes.
- (xiii) The following measuring transmitter features shall be provided as a minimum; additional requirements may be stated elsewhere in the Specification:
 - 1) Pulsed dc field excitation;
 - 2) Scalable pulse output for integration counter drive;

- 3) Capability of bi-directional measurement with differing forward and reverse ranges and with local and remote indication of flow reversal;
- 4) Contact operation at a programmable measured value;
- 5) Integral display of flow and integrated quantity;
- 6) Galvanic isolation between each output circuit and between the electrode circuit and output circuit;
- 7) Output circuit isolation from earth within the instrument but suitable for earthing at any point in the external circuit;
- 8) Key entry for basic parameters;
- 9) Commissioning and re-scaling to require no special programming knowledge;
- 10) Empty pipe detection using physical electrode;
- 11) Adjustable low flow cut-off;
- 12) Self-diagnosis;
- 13) Continuously adjustable velocity and flow range settings; and
- 14) Terminals accommodated in a compartment separate from electronic components.

(b) Performance

- (i) Unless an exception is provided in Particular Specifications P6 ICA, the following performance shall be available as a minimum; superior standards shall be met where stated elsewhere in the Specification or where so required by the metering and/or control requirements.

Flow as % of full scale	Maximum error
30 to 100%	±0.25 % of measured value
10 to 30%	±0.5 % of measured value
5 to 10%	±0.5 % of measured value
Repeatability	Better than ±0.2%

- (ii) Electromagnetic flowmeters shall perform continuous self-checking of the sensor and transmitter to ensure a higher accuracy & long term performance.
- (iii) Measuring transmitters shall be interchangeable with those of any other electromagnetic flowmeter of the same design, subject to the setting of calibration parameters for the associated sensor.
- (iv) Plant-mounted measuring transmitter enclosures shall have a degree of protection of not less than IP 65. Measuring transmitters installed within a panel located in a building may be of the rack-mounted type having a degree of protection not less than IP 20. Measuring transmitters shall not be located in flowmeter chambers or areas subject to flooding.

- (v) Electromagnetic flowmeters shall be supplied with a factory calibration certificate for a least a 5-point calibration and it shall be traceable to an accredited measuring standard. The following methods shall be used for factory calibration:
 - 1) BS EN 24185:1993, ISO 4185-1980: Measurement of fluid flow in closed conduits and weighing method;
 - 2) BS EN ISO 8316:1997: Method of liquid flow in closed conduits. Method of collection of the liquid in volumetric tank; and
 - 3) BS EN 29104:1993, ISO 9104:1991: Measurement of fluid flow in closed conduits. Methods of evaluating the performance of electromagnetic flowmeters for liquids.

G6.12.21 Thermal Flow Switches

- (a) Thermal flow switches shall have an adjustable switch-point and shall operate for either a high or low alarm condition.
- (b) The operation delay time shall be less than 20 seconds and the switching hysteresis shall be less than 10% of the selected range.
- (c) Wetted materials shall be suitable for the process fluid which may be liquid or gas. Special attention shall be paid to galvanic corrosion caused by parasitic currents in the measuring tip.
- (d) The flow switch shall have an enclosure to IP65 or better.
- (e) The flow switch shall have indicators for power and switch status which shall be visible with all covers closed.
- (f) Thermal flow switches shall be 24V dc powered, 4-wire devices. Outputs may be PNP or relay contact.
- (g) State of the output upon internally detected failure shall be configurable. Status of the output shall be indicated by means of visible LED or similar.

G6.12.22 Vortex Meters

- (a) Vortex meters shall comprise a transmitter unit and an in-line sensor factory installed into a spool piece. The transmitter may be attached to the sensor in the factory to form a one piece unit. Units with a separate power supply/signal demodulator shall also be acceptable. The probe-transmitter combination shall be selected and calibrated to suit the particular gas to be measured. The vortex flowmeter shall provide a 4-20 mA HART output signal proportional to the volumetric flow and a pulse output for totalisation. The transmitter shall provide local indication and calibration facilities.
- (b) Vortex meter sensors and spool pieces shall be constructed of stainless steel and suitable for the application and environment and shall be hermetically sealed to prevent the ingress of liquids and gases. The enclosure shall be IP65 or better.
- (c) Vortex meters shall be installed such that upstream and downstream straight pipe lengths are not less than those recommended by the manufacturer. Vortex meters shall be suitable for intrinsically safe installations where specified.

- (d) At Reynolds numbers over 20,000, the vortex meter shall have an accuracy equal to or better than $\pm 0.75\%$ of reading for liquids and $\pm 1.0\%$ of reading for gases and the repeatability shall be $\pm 0.2\%$ of rate or better. For Reynolds numbers between 4,000 and 20,000 the above accuracy and repeatability limits shall apply as percentages of full scale deflection.

G6.12.23 Thermal Flow / Mass Flow Meters

- (a) Thermal dispersion flow meters shall comprise a transmitter unit and a sensor probe for insertion into the process line, and shall be adjusted to suit the particular gas to be measured. Factory calibrations shall be performed to traceable standards.
- (b) The instrument shall be able to compensate deviations in readings caused by moisture in the process stream.
- (c) The flowmeter shall provide a 4-20 mA HART output signal proportional to the mass flow, and a pulsed output to give totalised flow. The transmitter shall provide local indication.
- (d) Flow meter probes shall be suitable for the application and environment and shall be hermetically sealed to prevent the ingress of liquids and gases.
- (e) Probe assemblies shall be complete with facilities including gate valves and packing glands to permit their removal from a running plant, at full line pressure without allowing any escape of media. i.e. a "hot tap" type fitting. The transmitter unit shall have an enclosure to IP65 or better. Thermal mass flow meters shall be suitable for intrinsically safe installations where specified.
- (f) Thermal dispersion flowmeter shall be of retractable type with an isolation valve to facilitate easy maintenance.
- (g) The flow meter shall have an overall accuracy equal to or better than $\pm 3\%$ of mass flow reading.
- (h) Thermal dispersion flow meters are not preferred for foul air service for the following reasons:
 - (i) Moisture effects; and
 - (ii) Fouling due to greasy air.

G6.12.24 Hygrometers (Moisture and Dew Point Monitors)

- (a) Hygrometers shall comprise a probe assembly, amplifier-transmitter and indicator unit.
- (b) The probe assembly shall operate on the capacitance or surface acoustic wave principle and shall be suitable for measurement of the moisture content of air or process gas as specified. The sensor shall be resistant to contamination by any other gas or vapour likely to be present. The probe assembly shall be mounted in the gas sample pipework which shall be fitted with a pressure gauge or manometer. Process gas shall be discharged to an approved point.
- (c) The amplifier-transmitter and indicator unit shall be suitable for wall mounting or panel mounting as required. Wall mounting units shall have enclosures with a degree of protection of IP 65.

- (d) The indicator shall be calibrated with a dew point scale from -100 to -20 °C unless otherwise specified. Two high value alarm points shall be provided with indicator lights, alarm set point adjustment, lamp test button and voltage-free contacts for re-transmission of both conditions.
- (e) The transmitter shall produce a 4-20mA HART output signal proportional to the linear indicator scale.
- (f) The overall accuracy of the hygrometer shall be better than ± 3 °C dew point and the repeatability shall be better than ± 1.5 °C dew point.

G6.12.25 Turbidity Meters

- (a) Turbidity meters shall be continuous flow, continuous reading, online instruments using the nephelometric principle of measurement. The turbidity meters shall employ 90° LED light technology in accordance with DIN EN ISO 7027, which in turn converts it to an electrical signal for processing and display.
- (b) For low turbidity measurements (under 1NTU) laser light technology should be used. Any deviation from this shall require approved by the S.O.
- (c) The turbidity meter shall provide direct measurement turbidity reading in NTU or FNU.
- (d) The turbidity meter shall provide an auto-ranging digital display with automatic decimal point positioning to eliminate range setting and reading errors.
- (e) A 4 to 20mA HART output signal programmed to cover any part of the instrument range shall be provided. An instrument failure alarm contact shall be provided.
- (f) The turbidity meter should include a debubbling chamber to minimize the impact of bubbles in the sample.
- (g) The sensor shall be equipped with self-cleaning devices for removing deposits from the sensor. The cleaning time and intervals are controlled by means of the transmitter. Materials of the sensor wetted parts and cleaning device shall be fully compatible with the measured fluid.
- (h) A calibration/verification module shall be provided along with the turbidity meter for checking and verifying the turbidity sensor.
- (i) Enclosures for the turbidimeter and control unit shall provide a degree of protection IP 65.
- (j) The accuracy from a range of 0 to 20 NTU shall be ± 0.02 NTU or $\pm 1.0\%$ of reading whichever is greater and above 20 NTU overall accuracy shall be $\pm 5\%$ of reading or better. The repeatability shall be $\pm 1.0\%$ of reading or ± 0.003 NTU or better whichever is greater. The resolution shall be 0.001 NTU or better.

G6.12.26 Total Suspended Solids Analysers (Optical)

- (a) Total suspended solids (TSS) analysers shall utilise alternating four-beam light technology with two pulsed infrared sources and four detectors to measure the incident light at differing angles. The analyser shall provide color-independent, sensor contaminant build-up and temperature compensated suspended solids measurement.

- (b) Where concentrations of suspended solids are expected to exceed 10 g/l, microwave-based Total Solids Meters are preferred.
- (c) The analyser shall provide a linear 4-20 mA HART output signal that corresponds to a pre-selected TSS range. The analyser shall have a fault relay output.
- (d) The sensor shall be equipped with self-cleaning devices for removing deposits from the sensor. The cleaning time and intervals are controlled by means of the transmitter. Non-contacting means of self-cleaning devices for sensor is preferred.
- (e) The overall accuracy shall be $\pm 5.0\%$ of reading or $\pm 1\%$ of full scale, whichever is greater. The repeatability shall be $\pm 1.0\%$ of reading or $\pm 0.2\%$ of full scale, whichever is greater.
- (f) Enclosures for the transmitter or control unit for indoor use shall provide a degree of protection IP 65 or better.

G6.12.27 Particle Counters

- (a) Particle counters shall be continuous flow, batch counting, on line instruments using the single particle light interaction principle of measurement.
- (b) The particle counter shall have at least four 4-20 mA HART outputs, each representing the number of particles within a defined size range per unit volume of sample. Though the 4-20 mA output shall be maintained continuously, it's value shall be updated on a discrete basis depending on the sample flow and the configured sample size. These parameters shall be set such that an update is obtained at least every 1 minute and also such that the coincidence error does not exceed 2%. Both the size ranges and unit volume of sample fluid shall be configurable. An instrument failure alarm contact shall be provided.
- (c) Each instrument shall be provided with the manufacturers recommended bubble trap, sample flowrate rotameter, and separate sample isolation and flow adjustment valves. The sample supply to the instrument shall avoid the use of pumps. If pumping is found to be absolutely necessary, the sample pump shall be located downstream of the particle counter and shall be a gear type pump.
- (d) The particle counter shall be capable of reliably counting particles down to $1\mu\text{m}$. A typical configuration of the configuration of the particle counter would be to provide a separate count for particle numbers in each of the following ranges: $1\mu\text{m}$ to $2\mu\text{m}$, $>2\mu\text{m}$ but $\leq 5\mu\text{m}$, $>5\mu\text{m}$ but $\leq 10\mu\text{m}$, and $>10\mu\text{m}$.
- (e) The particle size analyser wetted surfaces shall be manufactured from materials suitable for use with the raw and filtered water quality described elsewhere in this document. Enclosures for the particle counter and control unit for indoor use shall provide a degree of protection IP 65.
- (f) The analysers shall require recalibration at an interval not less than 12 months throughout the working life of the unit. Calibration certificates shall be provided with each instrument before the units are installed on site and this shall confirm the instrument has satisfactorily passed the calibration procedure tests. Details of these tests are to be forwarded to the S.O. with the detailed design information. The instrument performance shall be maintained without manual intervention for a minimum period of 30 days.

G6.12.28 pH Meters

- (a) Each pH metering system shall comprise electrode assembly, preamplifier (if appropriate) and analyser transmitter.
- (b) The pH electrode assembly shall be of the same manufacture as the associated preamplifier and analyser and unless otherwise specified shall be of the flow-through type.
- (c) Electrode assemblies shall comprise a glass electrode, a silver chloride (AgCl) reference electrode and a resistance thermometer for temperature compensation in an immersion type housing.
- (d) The measuring electrode shall be of glass and shall be designed for resistance to breakage and low electrical resistance. The flow direction in flow-through electrode assemblies shall be such as to give a self-scouring action at the electrodes.
- (e) When not online, the measuring electrode shall be installed in a specifically designed measuring cell, manufactured by the same supplier of the instrument. Site manufactured cells are not permitted.
- (f) When installed in a panel, flowcells will have their own dedicated feed, including a flow control needle valve, rotameter, flow switch and dedicated drainage. No daisy-chain assemblies will be permitted.
- (g) Electrode assemblies shall be easily detachable from their cells for buffering. Flow-through electrode mountings shall be designed to accommodate a beaker for containing buffer solutions and two such beakers shall be supplied with each flow-through electrode assembly.
- (h) The preamplifier, if required, shall be:
 - (i) Integral with the electrode assembly; or
 - (ii) In a separate enclosure mounted in the vicinity of the electrode assembly.
- (i) The pH analyser when mounted externally to a panel shall be protected to IP 65 and shall be suitable for surface mounting. The analyser shall have an integral indicator and shall produce a 4-20 mA HART output which is linear to pH.
- (j) The analyser span shall be adjustable after installation and the range shall be easily varied to cover a minimum range of 2 to 12pH. Calibration adjustments shall be provided for each end of the measuring range.
- (k) The accuracy of the pH analyser shall be ± 0.05 pH or better and the repeatability shall be ± 0.01 pH or better. This performance shall be maintained without manual intervention for a minimum period of 30 days. The response time shall not exceed 15 seconds.
- (l) Controls shall be provided for:
 - (i) Automatic/manual temperature compensation; and
 - (ii) Calibration adjustment.

- (m) Each analyser shall be provided with two sets of fully adjustable relay contacts which shall be able to be set to operate anywhere in the range of the analyser. In addition, a fault relay shall provide indication of any probe or analyser fault.
- (n) Where specified, pH metering systems shall incorporate equipment for automatic pH control in conjunction with equipment specified elsewhere.

G6.12.29 Oxygen Reduction Potential (ORP) Meters

- (a) Each ORP metering system shall comprise electrode assembly, preamplifier (if appropriate) and analyser transmitter.
- (b) The ORP electrode assembly shall be of the same manufacture as the associated preamplifier and analyser and unless otherwise specified shall be of the flow-through type.
- (c) Electrode assemblies shall comprise a glass electrode, a silver chloride (AgCl) reference electrode and a resistance thermometer for temperature compensation in an immersion type housing.
- (d) When not online, the measuring electrode shall be installed in a specifically designed measuring cell, manufactured by the same supplier of the instrument. Site manufactured cells are not permitted.
- (e) When installed in a panel, flowcells will have their own dedicated feed, including a flow control needle valve, rotameter, flow switch and dedicated drainage. No daisy-chain assemblies will be permitted.
- (f) The measuring electrode shall be of glass and shall be designed for resistance to breakage and low electrical resistance. The flow direction in flow-through electrode assemblies shall be such as to give a self-scouring action at the electrodes.
- (g) Electrode assemblies shall be easily detachable from their cells for calibration. Flow-through electrode mountings shall be designed to accommodate a beaker for containing buffer solutions and two such beakers shall be supplied with each flow-through electrode assembly.
- (h) The preamplifier shall, according to the particular application be:
 - (i) Integral with the electrode assembly; or
 - (ii) In a separate enclosure mounted in the vicinity of the electrode assembly.
- (i) The ORP analyser when mounted externally to a panel shall be protected to IP 65 and shall be suitable for surface mounting. The analyser shall have an integral indicator and shall produce a 4-20 mA HART output which is linear to ORP.
- (j) The analyser span shall be adjustable after installation and the range shall be easily varied to cover a minimum range of 300 mV. Calibration adjustments shall be provided for each end of the measuring range.
- (k) The accuracy of the ORP analyser shall be ± 2 mV or better and the repeatability shall be ± 2 mV or better. This performance shall be maintained without manual intervention for a minimum period of 30 days. The response time shall not exceed 15 seconds.

- (l) Controls shall be provided for:
 - (i) Automatic/manual temperature compensation; and
 - (ii) Calibration adjustment.
- (m) Each analyser shall be provided with two sets of fully adjustable relay contacts which shall be able to be set to operate anywhere in the range of the analyser. In addition, a fault relay shall provide indication of any probe or analyser fault.

G6.12.30 On-line Total Organic Carbon Analysers

- (a) The total organic carbon (TOC) analyser complete with inorganic carbon remover where applicable shall be designed for automatic continuous on-line measurement of non-purgeable organic carbon content in the water sample. The analyser shall operate on the principle of inorganic carbon removal and two-stage advanced oxidation for high resistance to chlorides and calcium.
- (b) The TOC analyser shall comply with the following requirements:
 - (i) It shall consist of integrated on-line sampling system;
 - (ii) The measuring range shall be between 0.03 ppb and 50 ppm;
 - (iii) All TOC analysis operations shall be fully automated and shall not require user intervention. The time taken for TOC analysis shall be within 10 minutes or faster;
 - (iv) The accuracy of the analyser shall be $\pm 2\%$ or ± 0.5 ppb, whichever is greater;
 - (v) The analyser shall have a calibration stability of at least 12 months;
 - (vi) It shall use an USEPA-approved method;
 - (vii) The display shall consist of a LCD screen with intuitive menu-driven and touch screen interface for ease of use;
 - (viii) The analyser shall have an automatic reagent injection system;
 - (ix) The various components of the analyser shall be housed in a wall mounted panel/enclosure rated Ingress Protection (IP) 45 or better; and
 - (x) TOC analysers shall have a 4-20 mA output with HART if available. There shall also be a potential free contact to indicate alarms and faults.

G6.12.31 Portable Total Organic Carbon (TOC) Analysers

- (a) Portable Total Organic Carbon (TOC) analyser shall operate on either of the following principles:
 - (i) Chemical stripping of inorganic carbon followed by high temperature catalytic oxidation of organic carbon to CO₂ followed by measurement of CO₂ concentration using non-dispersive infra-red detection; and
 - (ii) Chemical stripping of inorganic carbon followed by ultraviolet promoted persulfate oxidation and non-dispersive infra-red detection.
- (b) The portable TOC analyser shall be ambient temperature compensated and shall be suitable for operation over a TOC range of 0 to 10 ppm TOC.

- (c) The portable TOC analyser shall have an on-line accuracy of $\pm 3\%$ FSD or better and an on-line repeatability of $\pm 2\%$ of FSD or better. The minimum detection limit shall be 0.05 ppm or better.
- (d) The portable TOC analyser shall be suitable for current and future anticipated salinity levels.
- (e) Response time measured from the time the sample enters the instrument until a reading is obtained and any cleaning sequence required before the next sample can be taken is completed shall be 10 minutes or better. This shall include time for sample preparation, removal of inorganic carbon, oxidation, transport of CO₂ gas to analysing unit.
- (f) The portable TOC analyser shall be provided with an internal reagent packs. The internal reagent packs shall be able to be installed quickly within minutes and shall be able to provide three to six months of uninterrupted service.
- (g) TOC analysers shall have a 4-20 mA HART output that is proportional to total organic carbon concentration in the sample as well as selectable output for TOC or resistivity or temperature.
- (h) Portable TOC analysers shall have a high level communication port, able to extract data, calibration reports, etc.
- (i) The alarm outputs for Hi-Alarm, Lo-Alarm shall be provided.
- (j) The analyser shall be provided with a back-lit LCD display showing display of TOC, resistivity, temperature and operation/ error indications.
- (k) The power supply requirement shall be 230 Vac, 50 Hz.
- (l) Portable TOC analyser shall be provided with a RS-485 port for network communication capability, RS-232 ports for data acquisition, printer and diagnostic and a USB port that allows data transfer from analyser to a USB flash memory drive without interrupting analysis.

G6.12.32 Dissolved Oxygen Meters

- (a) The dissolved oxygen (DO) analyser shall employ optical technology except for industrial waste treatment applications which shall use membrane-type electrochemical sensors.
- (b) The dissolved oxygen meter shall incorporate automatic temperature, altitude and salinity compensation.
- (c) A cleaning system shall be provided for automatic in-situ cleaning of the sensor head.
- (d) The analyser when mounted externally to a panel shall be protected to IP 65 and shall be suitable for surface mounting. The analyser shall have an integral indicator and shall produce a 4 to 20 mA HART output which is linear to percent saturation of dissolved oxygen.
- (e) The analyser shall have a reasurment range of 0 to 20 mg/l.

- (f) The accuracy of the dissolved oxygen meter shall be $\pm 1\%$ saturation or better and the repeatability shall be $\pm 1\%$ saturation or better. This performance shall be maintained without manual intervention for a minimum period of 30 days.
- (g) The response time T_{90} shall be as short as possible.
- (h) A fault relay shall provide indication of any probe or analyser fault.
- (i) The DO measurement of the DO analyser shall not be interfered with H_2S , pH, K^{+1} , Na^{+1} , Mg^{+2} , Ca^{+2} , NH_4^{+1} , Al^{+3} , Pb^{+2} , Cd^{+2} , Zn^{+2} , Cr (total), Fe^{+2} , Fe^{+3} , Mn^{+2} , Cu^{+2} , Ni^{+2} , Co^{+2} , CN^{-1} , NO_3^{-1} , SO_4^{-2} , S^{-2} , PO_4^{-3} , Cl^{-1} , anion active tensides or crude oils.

G6.12.33 Residual Ozone Meters

- (a) Residual ozone meters shall comprise sensors, analyser-transmitters and indicator units. These shall be arranged for continuous monitoring of the residual ozone in samples delivered to the measuring cells from selected locations.
- (b) The sensor shall be of the amperometric type with plastic through-flow body, gas-permeable membrane, gold cathode, silver anode and buffered salt electrolyte. Reduction of the ozone at the cathode shall produce a current proportional to the ozone concentration in the water. The life of the electrolyte shall be at least six months. The sensor shall incorporate a thermistor for automatic temperature compensation over the range 0 to 50 °C and automatic pressure compensation over the range 100-550 kPa, a means of maintaining the electrodes free from fouling.
- (c) Unless otherwise specified, means shall be provided of detecting that sample flow is healthy. Voltage-free contacts shall be provided for remote signalling of high/low reading and instrument fault conditions.
- (d) The analyser-transmitter shall be of the surface mounting type, protected to IP 65 or better. The unit shall incorporate a liquid crystal display, a battery to maintain the cell polarising voltage in the event of power supply interruption, built-in diagnostics and simulator system with test switch.
- (e) The transmitter shall produce an isolated 4-20mA HART signal proportional to residual ozone in the range 0-1 mg/l unless otherwise specified. Two sets of voltage-free changeover contacts shall be provided for re-transmission of high and low alarm conditions.
- (f) The overall accuracy of the residual ozone meter shall be better than $\pm 3\%$ of fsd and repeatability shall be better than ± 0.03 mg/l. Resolution shall be 0.001 ppm ozone or better. The meter performance shall be maintained without manual intervention for a minimum period of 30 days.

G6.12.34 Residual Chlorine Meters (Non Multi Sensor Validation System)

- (a) The residual chlorine analyser shall comprise measuring flow-cell, transmitters and/or indicator units. These shall be arranged for continuous monitoring of the residual in samples delivered to measuring cells from selected locations.

- (b) The analyser sensor shall be of a membraned amperometric chlorine sensor and shall be enhanced with a third pH reference electrode which eliminates zero drift. The sensor design shall eliminate the requirement of external pH compensation, and the analyser shall be of a reagent-free type.
- (c) The materials of construction and sealing of sensor and flow-cell shall be compatible with the sample fluid. The flow-cell shall incorporate means of maintaining the electrodes free from fouling.
- (d) The sensor shall be capable of measuring both free available chlorine and total chlorine.
- (e) Automatic temperature compensation shall be provided for sample and ambient temperatures ranging between 0 - 50°C. The analyser shall be capable of operating in a pH range of pH 4 up to pH 9 without affecting measurement accuracy.
- (f) The transmitter shall incorporate an integral indicator and produce a 4-20 mA HART signal proportional to residual. The unit shall be housed in a surface mounting enclosure protected to IP 65. The fascia shall have indicator lights for high and low alarm conditions, alarm set point adjustment, lamp test facility and volt-free changeover contacts for re-transmission of high and low alarm conditions.
- (g) Unless otherwise specified, a means shall be provided of detecting that sample flow is healthy. Volt free contacts shall be provided for remote signalling of these conditions and any other instrument fault condition.
- (h) The measurement range shall be 0.01-10 mg/l with a minimum resolution of 0.01 mg/l. The overall accuracy of the residual chlorine meter shall be $\pm 3\%$ of the reference test (using DPD) or better and the repeatability shall be $\pm 5\%$ of full scale at stable temperature and pH. The analyser performance shall be maintained without manual intervention for a minimum period of 30 days.
- (i) Maintenance intervals for the membrane cap shall be greater than 6 months, and replacement intervals of electrolyte shall be greater than 3 months.

G6.12.35 Residual Chlorine Meters (Multi Sensor Validation System)

- (a) As shown in the Drawings Multi Sensor Validation (MSV) type of Residual Chlorine Meter shall be provided. The MSV Residual Chlorine meter shall be from a reputable manufacturer. It shall be submitted to S.O. for approval before considered acceptable.
- (b) MSV Residual Chlorine Meter shall comprise three cells, triplication controller and indicator units. It shall be arranged for continuous comparing and averaging of the residual in samples delivered to cells from selected locations. The controller shall analyse the three incoming signals and displays an average reading.
- (c) MSV Residual Chlorine Meter shall display error messages for conditions as follows:
 - (i) Level 1 alarm shall be triggered if any one of the measured signals deviated from the others by more than the preset tolerance band. The deviated signal shall be excluded from computing the average output signal; and

- (ii) Level 2 alarm shall be triggered if all three measured signals spread apart beyond the preset tolerance band. The output signal is either the last healthy average or a preset default (user selectable).
- (d) Each measuring cell shall be an amperometric instrument designed to generate signal proportional to the residual chlorine in the sample passed through the cell. Where necessary, each cell shall incorporate the facility for addition of a buffer solution to minimise the effect of pH variations on cell output. The materials of construction and sealing shall be compatible with the sample fluid. Cells shall incorporate means of maintaining the electrodes free from fouling.
- (e) Each cell shall be capable of measuring both free available chlorine and total chlorine. Where required, each cell shall be completed with buffer pump, suction & discharge tubing. The cell's reagents are fed from a reservoir through a rotary valve. The reservoir is filled through the reagent pumps from a floor mounted container. All components are accessible from the front of the unit to permit visual inspection of solution levels.
- (f) Where reagents are used, they shall require replenishment not more frequently than once per 30 days. The reagent feed shall be stopped automatically in the event that sample flow is lost.
- (g) Unless otherwise specified, a means shall be provided of detecting that sample flow is healthy and, where applicable, that a reagent supply is available. Volt free contacts shall be provided for remote signalling of these conditions and any other instrument fault condition.
- (h) Automatic temperature compensation shall be provided for sample and ambient temperatures ranging between 0 – 50 Deg C. Flow switch shall be provided and mounted on the inlet of the sampling unit to detect the loss of sample to the sampling unit. Low flow alarm shall be triggered when no sample flow.
- (i) MSV Residual Chlorine Meter shall incorporate an integral indicator and produce a 4-20 mA HART signal proportional to residual chlorine. The average and individual cell reading shall be displayed on four-digital LCD. The measuring range shall be from 0 to 5ppm. The minimum and maximum span's values shall be entered from the keyboard.
- (j) MSV Residual Chlorine Meter shall have microprocessor to monitor cells measurement, activate warning for signals deviation, self-diagnostic function via keyboard and record output reading.

MSV Residual Chlorine Meter shall be housed in an IP66 clear fronted plastic surface mounted enclosure with plug in power supply module, processor module and alarm module. The fascia shall have indicator lights for Low-Low, Low, High and High-High alarm conditions, alarm set point adjustment, lamp test facility and volt-free contacts.
- (k) The overall accuracy of the residual chlorine meter shall be $\pm 2\%$ of full scale or better at fixed temperature and pH. Subject to replenishment of reagents, where applicable, the meter performance shall be maintained without manual intervention for a minimum period of 30 days.

G6.12.36 Ammonia/Monochloramine Analysers

- (a) The Low Range ammonia/monochloramine analyser shall be a microprocessor-controlled colorimetric analyser. The measurement ranges shall be 0.02 to 2.0 mg/L as nitrogen for ammonia residual measurement or 0.1 to 10.0 mg/L as chlorine for chloramine measurement.
- (b) The analyser accuracy shall be $\pm 5\%$ or ± 0.02 mg/L (as N), whichever is greater.
- (c) The analyser shall display data in a numeric or graphical format. The analyser shall be capable of automatic calibration, cleaning and self-priming. The analyser shall be capable of grab-sample analysis without interrupting continuous sample flow to the analyser.
- (d) The analyser shall be provided with a transmitter that includes a minimum of an isolated 4-20 mA HART output. Transmitter output span shall be user-adjustable over the entire span of the analyser. Two non-powered SPDT relays shall be provided for internal alarms. Two relay contacts shall be rated for 5A at 240 V ac. Alarms may be programmed for sample concentration alarms, analyser system warning and analyser system shutdown.
- (e) The transmitter shall be supplied with an appropriate length of signal cable for connecting the transmitter to the sensor.
- (f) The analyser components shall be housed in a stainless steel or cast aluminium or fiberglass reinforced plastic IP65 enclosure designed for bench-, wall- or panel-mounting. The analyser shall be warranted for one full year against defects in materials and workmanship and shall include a 30-day supply of standards and reagents.

G6.12.37 Ammonia Analysers (GSE & ISE)

- (a) The ammonia analyser shall be either a gas sensing electrode (GSE) type or an ion selective electrode (ISE) type as specified in the instrument list.
- (b) ISE type analysers shall not be used if interference from the expected concentration of chloride or other ions will impact the ammonia measurement by more than 2%.
- (c) The measurement ranges shall be of 0.05 to 20.0 mg/L as nitrogen.
- (d) The analyser accuracy shall be $\pm 3\%$ or ± 0.05 mg/L (as N), whichever is greater.
- (e) The analyser shall display data in a numeric or graphical format. The analyser shall be capable of automatic calibration and cleaning.
- (f) The analyser shall be equipped with self-diagnostic routines.
- (g) The measurement interval shall be 5 to 120 minutes, adjustable. The response time shall be less than 5 minutes (T90) including sample preparation.

The analyser shall be provided with a transmitter that includes a minimum of an isolated 4-20 mA HART output. Transmitter output span shall be user-adjustable over the entire span of the analyser. Two non-powered SPDT relays shall be provided for internal alarms. Two relay contacts shall be rated for 5A at 240 V ac. Alarms may be programmed for sample concentration alarms, analyser system warning and analyser system shutdown.

- (h) The transmitter shall be supplied with an appropriate length of signal cable for connecting the transmitter to the sensor.
- (i) The analyser components shall be housed in a stainless steel or cast aluminium or fiberglass reinforced plastic IP65 enclosure designed for bench-, wall- or panel-mounting. The analyser shall be warranted for one full year against defects in materials and workmanship and shall include a 3-month supply of reagents.
- (j) The analyser shall be equipped with a sample preparation, i.e. sampling system as follows when specified:
 - (i) A filtration system from the analyser manufacturer to reject solids in the samples or reagents;
 - (ii) All control components of the sampling system shall be housed and heated in an IP55-rated enclosure, suitable for outdoor installation; and
 - (iii) The sampling system shall provide two programmable and potential-free contacts (240VAC @ 3A) to detect high or low flow passing through the filter media.

G6.12.38 Nitrite/Nitrate Analysers

- (a) The nitrite/nitrate analyser shall be based on the detection and analysis of the natural light absorbance signature for nitrite and nitrate in the ultraviolet (UV) wavelength range. Multiple wavelength pattern recognition techniques shall be employed to extract the contribution made by any single parameter to the total light absorbance signature of the solution and allows nitrite and/or nitrate to be detected as an individual parameter.
- (b) The analyser shall be able to measure the concentration of nitrite (NO₂ as N) and nitrate (NO₃ as N) simultaneously.
- (c) The analyser shall employ a drift-free, dual beam spectrometer with optical resolution better than 1nm.
- (d) The analyser shall be equipped with automatic cleaning mechanism.
- (e) The analyser shall be provided with a transmitter that includes a minimum of an isolated 4-20 mA HART output. Transmitter output span shall be user-adjustable over the entire span of the analyser.
- (f) The transmitter shall be supplied with an appropriate length of signal cable for connecting the transmitter to the sensor.
- (g) The analyser components shall be housed in a stainless steel or cast aluminium or fiberglass reinforced plastic IP65 enclosure designed for bench-, wall- or panel-mounting. The sensor shall be of IP68.
- (h) The analyser shall be suitable for wastewater treatment plant applications.

G6.12.39 Low Range Hardness Analysers

- (a) The method of measuring low range hardness shall be a microprocessor-controlled process analyser designed to use an EPA-approved calmagite chemistry method to monitor low range hardness colorimetrically.
- (b) The measuring range shall be 0.5 to 10 mg/l as CaCO₃.

- (c) The analyser accuracy shall be $\pm 5\%$ or ± 0.05 mg/L (as CaCO_3), whichever is greater.
- (d) Repeatability shall be $\pm 3\%$ or ± 0.03 mg/L (as CaCO_3), whichever is greater.
- (e) The analyser shall have a response time <5 minutes to 90% of reading.
- (f) The analyser shall display data in a numeric or graphical format. The analyser shall be capable of automatic calibration, cleaning and self-priming.
- (g) Sample shall be continuously purged to assure fresh sample to the analyser and reduce analysis lag time.
- (h) The analyser shall be provided with a transmitter that includes a minimum of an isolated 4-20 mA HART output. Transmitter output span shall be user-adjustable over the entire span of the analyser. Two non-powered SPDT relays shall be provided for internal alarms. Two relay contacts shall be rated for 5A at 240 V ac. Alarms may be programmed for sample concentration alarms, analyser system warning and analyser system shutdown.
- (i) The transmitter shall be supplied with an appropriate length of signal cable for connecting the transmitter to the sensor.
- (j) The analyser components shall be housed in a stainless steel or cast aluminium or fiberglass reinforced plastic IP66 enclosure designed for bench-, wall- or panel-mounting.
- (k) The analyser shall be warranted for one full year against defects in materials and workmanship and shall include a 3-month supply of reagents.

G6.12.40 High Range Hardness Analysers

- (a) The method of measuring high range hardness shall be a microprocessor-controlled process analyser designed to use an EPA-approved calmagite chemistry method to monitor low range hardness colorimetrically.
- (b) The measuring range shall be 10 to 1,000 mg/l as CaCO_3 .
- (c) The analyser accuracy shall be $\pm 5\%$ or ± 2 mg/L (as CaCO_3), whichever is greater.
- (d) Repeatability shall be $\pm 5\%$ or ± 5 mg/L (as CaCO_3), whichever is greater.
- (e) The analyser shall have a response time <17 minutes to 90% of reading.
- (f) The analyser shall display data in a numeric or graphical format. The analyser shall be capable of automatic calibration, cleaning and self-priming.
- (g) Sample shall be continuously purged to assure fresh sample to the analyser and reduce analysis lag time.
- (h) The analyser shall be provided with a transmitter that includes a minimum of an isolated 4-20 mA HART output. Transmitter output span shall be user-adjustable over the entire span of the analyser. Two non-powered SPDT relays shall be provided for internal alarms. Two relay contacts shall be rated for 5A at 240 V ac. Alarms may be programmed for sample concentration alarms, analyser system warning and analyser system shutdown.
- (i) The transmitter shall be supplied with an appropriate length of signal cable for connecting the transmitter to the sensor.

- (j) The analyser components shall be housed in a stainless steel or cast aluminium or fiberglass reinforced plastic IP66 enclosure designed for bench-, wall- or panel-mounting.
- (k) The analyser shall be warranted for one full year against defects in materials and workmanship and shall include a 3-month supply of reagents.

G6.12.41 Phosphate Analysers

- (a) The phosphate analyser shall be a microprocessor-controlled colorimetric analyser. The measurement ranges shall be 0.05 to 15.0 mg/L PO₄ as P.
- (b) The analyser accuracy shall be $\pm 2\%$ or ± 0.05 mg/L, whichever is greater.
- (c) The analyser shall display data in a numeric or graphical format. The analyser shall be capable of automatic zero-calibration and cleaning.
- (d) The analyser shall be provided with a transmitter that includes a minimum of an isolated 4-20 mA HART output. Transmitter output span shall be user-adjustable over the entire span of the analyser. Two non-powered SPDT relays shall be provided for internal alarms. Two relay contacts shall be rated for 5A at 240 V ac. Alarms may be programmed for sample concentration alarms, analyser system warning and analyser system shutdown.
- (e) The transmitter shall be supplied with an appropriate length of signal cable for connecting the transmitter to the sensor.
- (f) The analyser components shall be housed in a stainless steel or cast aluminium or fiberglass reinforced plastic IP65 enclosure designed for bench-, wall- or panel-mounting. The analyser shall be warranted for one full year against defects in materials and workmanship and shall include a 3-month supply of reagents.

G6.12.42 Conductivity Meters

- (a) The conductivity measurement system shall comprise conductivity cell, conductivity indicator, interconnecting cable etc.
- (b) The conductivity constant of the probe shall be adequate to the measuring range, from ultra low, low, and high conductivity.
- (c) Conductivity cells shall be of the same manufacturer as the associated indicator and shall be of the insertion type with facilities including gate valves and packing glands to permit their removal from a running plant, at full line pressure without allowing any escape of liquid i.e. "hot tap" type fitting. The cable entry fitting shall be waterproof.
- (d) The mounting positions for electrode cells shall ensure good circulation and that the measurement is representative of stream.
- (e) If flow cells are used, they shall be designed to completely avoid any reading disturbance created by bubble accumulations on the cell.
- (f) Conductivity transmitters shall have:
 - (i) A 4-20 mA HART output proportional to conductivity;
 - (ii) A built-in check circuit to perform calibration checks;

- (iii) Automatic temperature compensation;
- (iv) Accuracy of $\pm 1.0\%$ of fsd or better, repeatability of $\pm 0.5\%$ of fsd or better; and
- (v) A relay alarm output to indicate cell fouling other serious fault condition.

G6.12.43 Fluoride Meters/Monitors

- (a) Fluoride monitors shall employ the "selective ion" method to continuously and automatically monitor the level of fluoride in the sample. The monitor shall provide an isolated analogue 4-20 mA HART output signal which is proportional to the amount of fluoride in the sample.
- (b) The performance of the instrument should meet the following criteria:
 - (i) Accuracy or trueness not exceeding 10% of the result or 0.15mgF/l (whichever is the greater), at the 95% confidence level;
 - (ii) Precision or total standard deviation (under repeatability conditions) not exceeding 10% of the result or 0.15mgF/l (whichever is greater); and
 - (iii) Limit of detection not greater than 0.20mgF/l.
- (c) The instrument should be subject to an automatic standardisation with user configurable frequency. Standardisation must be at two concentrations spanning the target dose (normally 1.0mgF/l and within the range 0.5 to 1.5mgF/l. Where the instrument is used to control (or trim) the amount of chemical dosed, the lower concentration of the standardisation should be close to the target dose.
- (d) The instrument should allow the manual initiation of a standardisation.
- (e) High and low alarms shall be available as volt-free contacts for both local and remote alarm indication. Alarm set points shall be adjustable over the full range of the instrument. Additional volt-free contacts shall be provided for "system failed" indication. Alarm for loss of water sample to the instrument shall also be generated.
- (f) To ensure accurate measurement values, the instrument shall use a form of two point calibration which applies zero drift and slope correction to the sensor output. Automatic calibration shall be provided.
- (g) The monitors shall be housed in a suitable enclosure to protect equipment against the environment and shall be protected to IP 65 or better.
 - (i) Range shall be 0 to 1.5 ppm;
 - (ii) Accuracy shall be $\pm 5\%$ or better of full scale;
 - (iii) Repeatability shall be $\pm 2\%$ of reading or better; and
 - (iv) Local indication shall be provided.

G6.12.44 Ammonia and Chlorine Leak Detectors

- (a) Leak detection equipment shall comprise detector heads and remote mounted transmitters/monitors.
- (b) The detector heads shall be of the bimetallic amperometric or solid state types designed for stability and minimum maintenance. The leak detection system shall operate on a continuous basis with a maximum response time to low level leaks of 5 seconds. They shall be wall-mounted and reasonably equidistant from potential sources of ammonia or chlorine leakage as appropriate.
- (c) Each detector head shall be connected electrically to its associated transmitter/monitor. Multi-point "sniffer" type pumped sampling systems with a common detector head shall not be acceptable.
- (d) The ammonia detector shall not respond to a leakage of chlorine or sulphur dioxide.
- (e) The chlorine detector shall not respond to a leakage of ammonia or sulphur dioxide.
- (f) The cells shall be suitable for operation in an indoor environment with a temperature maintained above freezing point (0°C).
- (g) The monitoring units shall have wall mounting enclosures protected to at least IP 54 or shall be panel mounted. The units shall not be mounted in the gas hazard area. They shall incorporate solid state circuit cards of the plug-in type. The units shall have self-checking circuitry and detector failure alarm indication, the latter with provision for re-transmission. They shall be equipped with a back-up battery power supply to maintain unit operation for a minimum period of four hours in the event of mains power failure. In normal operation, the batteries shall be trickle-charged.
- (h) Leakage alarms shall be adjustable from 5% to 100% of the instrument range to facilitate setting of the following trip levels:
 - (i) Ammonia
 - low level 25 ml/m³
 - high level 35 ml/m³
 - (ii) Chlorine
 - low level 0.5 ml/m³
 - high level 1.0 ml/m³
- (i) The monitors shall have indicator lights for both alarm levels and at least two sets of volt-free changeover contacts for re-transmission of the alarms to:-
 - (i) Ventilation system controls (inhibit action) and external warning lamp; and
 - (ii) Remote monitoring system.
- (j) Local alarm indication and contact position shall be maintained until an alarm reset button is activated at the monitor.

- (k) The monitors shall incorporate a test button to facilitate the checking of satisfactory operation of the electronic circuits and alarms. Any consumable items that may be required for testing or re-calibration of the leak detection system shall be supplied in a sufficient quantity to last at least twelve months from take-over.

G6.12.45 Carbon Dioxide Leak Detectors

- (a) Carbon dioxide in air shall be measured by the non-dispersive infra-red method at a wave length specific to carbon dioxide. The instrument shall have a range of measurement greater than 0-5 l/m³ and shall be provided with two adjustable alarm setpoints within the operating range, and an instrument malfunction alarm.
- (b) The accuracy of the measurement shall be better than $\pm 5\%$ of the full scale deflection and the response time of the instrument shall be not greater than 20sec.
- (c) The instrument shall have an in-built sample pump capable of withdrawing a sample from up to 10 m away.
- (d) The instrument shall be suitable for wall mounting and shall be provided with local analogue or digital indication and facilities for providing 4-20 mA signals for remote indication of the carbon dioxide concentration. The instrument shall also incorporate relays providing volt-free contacts to transmit alarms to remote panels.
- (e) The instrument shall incorporate test buttons on the front of the panel facilitating quick checks of satisfactory operation of the electronic circuits. These circuits shall include input mains filter and high noise rejection components to afford protection from transients and to make the units suitable for the environment.

G6.12.46 Hydrogen Sulfide (H₂S) Gas Concentration Analysers

- (a) Hydrogen sulfide concentration analysers shall use the measurement method specified in the Instrument List. If none is specified then either chemical impregnated paper-tape colorimetry or dispersive UV-Vis absorbance spectrophotometry measurement techniques shall be used.
- (b) For applications requiring measurement below 1.5 ppm (1500 ppb), only chemical impregnated paper-tape colorimetry shall be used.
- (c) The detection limit and the response time shall be:
 - (i) UV-type
 - 1) 1ppm; and
 - 2) not greater than 20 sec.

- (ii) Paper tape type:
 - 1) Depending on selected tape-cassette either better than ± 100 ppb (0.1 ppm) or better than ± 10 ppb (0.01 ppm); and
 - 2) Not greater than 60 sec.
- (d) The analyser system shall have a sample pump capable of withdrawing a sample from and returning the exhaust to the process pipe line.
- (e) The gas sample shall be pretreated to ensure humidity and particulates are within the analyser manufacturers specified optimum range.
- (f) The design of the flow cell and analyser shall enable the analyser to be maintained without exposure to the sample gas stream when the analyser cabinet is opened for maintenance.
- (g) Where analyser configuration is of multiple sample stream switching, the sample gas must be continuously flowing in each sample lines to prevent stagnant flow and memory effect of hydrogen sulphide in the sample lines. The response time for each line shall be within 1 minute lag time. All sample streams shall return back to the process pipe line after the measurement or bypass.
- (h) The analyser shall be able to compensate for any pressure variation in the sample gas.
- (i) The sample lines shall be PFA or passivated metal tubes to prevent memory effect of hydrogen sulphide in the sample line.
- (j) The instrument shall be suitable for wall mounting and shall be provided with local analogue or digital indication and facilities for providing 4-20 mA signals, and preferably 4-20mA HART, for remote indication of H_2S concentration. The instrument shall also incorporate relays providing volt-free contacts to transmit alarms to remote panels.
- (k) The controller enclosure shall be of IP65 or better.
- (l) When multi-stream measurements are required, a gas sampling unit shall be provided for as follows:
 - (i) Active gas sampling pump module in conjunction with gas sensor and transmitter. Pump sampling module shall provide position indication of operation with flow switch;
 - (ii) Capable of scheduled sharing up to 8 sample points;
 - (iii) Provide all pump assemblies and valving to allow multiple gas sampling as specified and purging as required;
 - (iv) All components shall be sized to allow sampling of points up to 100 meters from the sample pump at flows and pressures compatible with the sensor; and
 - (v) Samples shall be conditioned so as to provide optimal life of the sensor. Conditioning shall include filtering and drying of the sample gas and draining as well.

G6.12.47 Hydrogen Leak Detectors

- (a) Hydrogen leak detectors shall comprise a detector cell or cells and an alarm unit.
- (b) The detector cells shall be strategically placed to detect the highest concentrations of the leaking gas with the sensor facing downwards. Where mounting is on a vertical wall a 90° mounting bracket shall be used.
- (c) The detector cell shall be of the electrocatalytic type or non-dispersive infra- red method at a wave length specific to Hydrogen and shall be completely self-contained including the sensor, electronics and cable connection facilities.
- (d) The detector housing material shall be stainless steel and the housing shall be certified as explosion proof for hydrogen gas areas to a reputable international standard. The certification of the detector shall be submitted to S.O. for approval before the instrument is used.
- (e) The alarm unit shall be provided in either a wall mounted enclosure or in a rack mounted system within a larger alarms panel. The alarm unit shall continuously monitor the detector head for changes due to a gas leak. The alarm unit shall be installed outside the hazardous area.
- (f) The alarm unit shall be provided with a continuous digital display of the gas level which shall be calibrated in percent LEL (lower explosive limit) and ranged from 0 to 100%. The unit shall provide two outputs for high and low alarms; the reference point shall be adjustable in the range 5 to 65% of full scale. The alarm unit shall also be provided with a unit failure alarm which shall generate an alarm on loss of signal from the sensor or power failure to the unit or an internal unit malfunction. Each alarm shall provide a separate volt free alarm relay output contact for connection to remote alarm systems.
- (g) The alarm unit shall provide an isolated 4-20mA HART analogue output signal linearly proportional to the gas level. The output shall be capable of driving into a minimum load of 200 ohms.
- (h) The alarms shall be displayed on the front of the alarm panel and shall operate in a normal alarm system mode with acknowledge and reset pushbutton for accepting and cancelling the alarm state. Automatic clearing of the alarm state is not acceptable.
- (i) The gas detection system range shall be 0-100%LEL with an accuracy of $\pm 5\%$ of full scale maintained within the full operating temperature range of the instrument. The operating temperature range of the detector cell shall be -40°C to +85°C and the operating range of the alarm unit shall be -5°C to +45°C unless the average environmental conditions in the area to which the instrument is to be installed exceed these limits, in which case the average environmental conditions shall apply. Response time of the detector system shall be less than 10 seconds for a step of 5% to 95%LEL change in gas level.
- (j) The power to the detection system installation shall be supplied from an uninterruptible power supply capable of powering all gas leak systems for a minimum of two hours after loss of mains power.
- (k) The gas detector system shall be set up to provide warning of hydrogen gas at sub-explosive levels.

G6.12.48 Ozone Gas Analysers

- (a) Ozone gas analysers shall be classified into the following types:
 - (i) Atmospheric monitors with a typical measurement range of 0-1.0 ppm by volume; and
 - (ii) Process gas analysers with typical measurement ranges of 0-1000 ppm up to 0-50000 ppm by volume.
- (b) The monitors and analysers shall operate on the principle of absorption of ultra-violet light. Monitors and analysers shall have a potential-free fault relay to signal of any internal fault. Enclosures shall be to IP65 or better.
- (c) Atmospheric monitors shall incorporate an integral dust filter, gas sample pump, flowmeter and diverter solenoid valve allowing the sample to pass directly to the measurement cell or indirectly via a catalytic converter to remove ozone from the sample for zero checking.
- (d) The ozone concentration shall be computed automatically from Beer's Law and corrected for temperature and pressure. The result shall be displayed in ppm of ozone by volume.
- (e) Unless otherwise specified, the range of measurement shall be 0-1.0 ppm with a minimum detectable limit of 0.002 ppm. Zero and span drift shall not exceed 1% of span per month. Linearity shall be ± 0.001 ppm or better. Resolution shall be 0.001 ppm or better.
- (f) The monitor shall have an isolated 4-20mA HART output proportional to ozone concentration.
- (g) When employed as a gas leak detector, a separate monitoring station shall be located away from the toxic gas hazard area. The monitoring station shall comprise an analogue repeater indicator and two trip amplifiers set to generate alarms at ozone concentrations of 0.1 ppm and 0.2 ppm. Indicator lights with lamp test button shall be provided for each alarm level and at least two sets of fail-safe voltage-free contacts for re-transmission of the alarms:
 - (i) To ventilation system controls (enabling action) and warning lights at each entry point to the area; and
 - (ii) To remote monitoring system.
- (h) Unless otherwise specified, the monitoring station shall be contained in an IP65 or better rated wall mounting enclosure located outside the main entry point to the hazard area. A fail-safe volt-free contact for instrument fault conditions shall be provided.
- (i) Process gas analysers shall incorporate an integral flow meter and diverter solenoid valve allowing either a feed gas sample (without ozone) or an ozone gas sample to pass to the measurement cell. The exhaust connection from the cell shall be connected to an approved discharge point which shall normally be the inlet to the vent ozone destructor.

- (j) The ozone concentration shall be computed automatically from Beer's Law. The zero point shall be adjusted automatically according to ultra-violet transmission through the feed gas reference sample. The span of the instrument shall be corrected for any electro-optical changes.
- (k) The range of measurement shall be selected according to the particular application. For ozone production control, the instrument display shall normally be calibrated in units of g/m³. Provision shall be made to hold the output signal delivered to the production control system when the analyser is performing a zero check. The output signal shall be 4-20mA HART proportional to ozone concentration.
- (l) The measurement accuracy shall be better than 3%, repeatability shall be ± 0.5 g/m³ or $\pm 1\%$ of the reading whichever is the greater, or better. Resolution shall be ± 0.1 g/m³ or better. Span and zero drift shall not exceed 0.5% per month.

G6.12.49 Oxygen Gas Analysers

- (a) Oxygen gas analysers should be capable of typical measurement range of 0-25% by volume.
- (b) Oxygen Analyser will contain electrochemical oxygen sensor which will be easy to dismantle for replacement and maintenance purposes. Monitors and analysers shall have a potential-free fault relay to signal of any internal fault. Enclosures shall be to IP65 or better.
- (c) Atmospheric monitors shall incorporate an integral dust filter, flowmeter and diverter solenoid valve allowing the sample to pass directly to the measurement cell.
- (d) Where required, the analyser shall be integrated with gas sample pump capable of extracting gas sample up to 10m away and returned back to the process line up to 10m away.
- (e) The sample probe with remote activated 24V DC blow back valves shall be provided at the sample take-off point to overcome dirty sample gas. The blow back signal shall be provided by remote panels.
- (f) Sample cooler shall be provided to remove moisture to avoid moisture interference for the NDIR analyser analysis.
- (g) When employed as a gas leak detector, a separate monitoring station shall be located away from the toxic gas hazard area. The monitoring station shall comprise an analogue repeater indicator and two trip amplifiers set to generate alarms at 19.5% and 23.5% oxygen level. Indicator lights with lamp test button shall be provided for each alarm level and at least two sets of fail-safe voltage-free contacts for re-transmission of the alarms:
 - (i) To ventilation system controls (enabling action) and warning lights at each entry point to the area; and
 - (ii) To remote monitoring system.
- (h) The monitor shall have an isolated 4-20mA HART output proportional to the measured oxygen concentration.

- (i) Unless otherwise specified, the monitoring station shall be contained in an IP65 or better rated wall mounting enclosure located outside the main entry point to the hazard area. A fail-safe volt-free contact for instrument fault conditions shall be provided.
- (j) Process gas analysers shall incorporate an integral flow meter and flow control valve allowing either a feed gas sample or an oxygen gas sample to pass to the measurement cell.
- (k) The range of measurement shall be selected according to the particular application. The instrument display shall normally be calibrated in % by volume. Provision shall be made to hold the output signal delivered to the production control system when the analyser is performing a zero check.
- (l) The measurement accuracy shall be better than $\pm 0.5\%$ vol, repeatability shall be $\pm 1\%$ of the reading, or better. Span and zero drift shall not exceed 3% per month.
- (m) The analyser shall provide with local indication for analogue and alarm status and able to provide 4-20mA HART signals for remote indication of the O₂ concentration.
- (n) The analyser shall also incorporate relays providing volt-free contacts to transmit alarms to remote panels.

G6.12.50 In-Situ Zirconium Oxide Cell Oxygen Gas Analysers

- (a) Where in-situ zirconium oxide cell method is specified, the oxygen gas analyser shall be in-situ sampling done by deflecting tube mounted on the process pipe or duct mating flange.
- (b) On-site calibration validation shall be performed from the transmitter. Calibration gases shall sent to the probe from a remote calibration system.
- (c) Oxygen transmitter shall be installed remotely from the probe, either on self-standing racks at ground level or on wall-mounted panels on platform. The transmitter shall supply power for heating probes to their operation temperature. Sensor signal shall be converted to oxygen % volume concentration on transmitter screen.
- (d) Calibration gas cylinder with stainless steel pressure regulator for automatic zeroing and span adjustment shall be provided.
- (e) The response time of the analyser shall be less than <7s for 90% of final value (from calibration gas inlet).
- (f) The analyser range shall be from 0-2% to 0-50% O₂ freely settable by vol% O₂ pitches.
- (g) The analyser shall provide with local indication for analogue and alarm status and able to provide 4-20mA HART signals for remote indication of the O₂ concentration. The analyser shall also incorporate relays providing volt-free contacts to transmit alarms to remote panels.
- (h) The analyser enclosure shall be IP 65 or better ingress protection. The analyser shall be able to operate in classified zone areas with relevant certificate.

G6.12.51 Online Algae Monitors

- (a) Algae monitors shall base on Chlorophyll Fluorescence or spectral fluorometry principle to continuously and automatically monitor the level of chlorophyll in the sample. The monitor shall provide an isolated analogue 4-20 mA HART output signal which is proportional to the amount of chlorophyll in the sample.
- (b) The analyser shall include automatic cleaning facilities for sensor chamber cleaning.
- (c) Maintenance interval of the analyser shall be greater than 7 days.
- (d) High and High-High alarms shall be available as volt-free contacts for both local and remote alarm indication. Alarm set points shall be adjustable over the full range of the instrument. Additional volt-free contacts shall be provided for "system failed" indication. Alarm for loss of water sample to the instrument shall also be generated.
- (e) The monitors shall be housed in a suitable enclosure to protect equipment against the environment and shall be protected to IP 54 or better.
- (f) The performance of the analyser should meet the following criteria:
 - (i) Range shall be 0 to 200 ug/l Chlorophyll a;
 - (ii) Accuracy shall be ± 0.2 ug/l or 3% of the reading, whichever is greater;
 - (iii) Repeatability shall be $\pm 3\%$ of reading or better;
 - (iv) Data storage shall be 20,000 points or more and equipped with serial connection for data export; and
 - (v) Local indication shall be provided.

G6.12.52 Floating Hydrocarbon Detectors

- (a) Floating hydrocarbon detectors shall be wave rider floating detector, capable of monitoring hydrocarbons and other organic solvents on water.
- (b) The analyser shall perform as per manufacturer's specification under the following operating conditions:
 - (i) Resolution: less than 1 mm of Hydrocarbon on Water;
 - (ii) Working Wave Height: Maximal 2 meters;
 - (iii) Tide Range: Unlimited;
 - (iv) Current: Up to 4 knots;
 - (v) Minimum liquid Depth: 30 cm;
 - (vi) Working Temperature: 0° C to 70° C;
 - (vii) Ambient Temperature: -40° C to 85° C; and
 - (viii) Survival Conditions: Extreme sea conditions.
- (c) Material Requirements
 - (i) Detector: 316 Stainless Steel, Hydrocarbon Resistant Polymers; and

- (ii) Wave Rider Buoy: Diameter 900 Mm, Height 300 Mm.
- (d) Dimensions
 - (i) Cable: 50 m Floating Cable with Detector;
 - (ii) Rider Buoy: Diameter 900 mm, Height 300 mm; and
 - (iii) Signal Processor Housing: NEMA 4 (IP-65).
- (e) Electrical Rating
 - (i) Wiring Connections: Barrier Terminal Block #1 8 AWG Maximum; and
 - (ii) Input Power Options: 24 VDC or Solar Powered.
- (f) Output Signals
 - (i) 4-20mA signal proportional to hydrocarbon thickness up to 1" (25mm) including Built-in Test feature;
 - (ii) Alarm Relay Contacts: SPDT with programmable response time, and three indicating lights; and
 - (iii) Air/Oil/Water Bar-Graph Display (20 Bars) of Hydrocarbon Thickness.
- (g) Certification Requirements
 - (i) Detector: Intrinsically Safe - CENELEC T4; BASEEFA; and
 - (ii) Signal Processor: Explosion Proof NEMA 7 (N7), Div.1 Class 1, Groups D.

G6.12.53 Silt Density Index Analysers

- (a) Silt Density Index (SDI) analysers shall automatically and consistently monitor the silt density index of Reverse Osmosis (RO) feed water.
- (b) The analyser shall be microprocessor controlled and fully automatic.
- (c) The analyser shall provide an isolated analogue 4-20 mA HART output signal which is proportional to the level of SDI in the sample.
- (d) The testing method shall comply with ASTM D 4189-07.
- (e) The analyser shall be designed with continuous sample flow and for permanent installation. The sampling system shall prevent stagnant conditions in the sample line. The analyser shall continuously allow a side stream of sample to flow through the analyser up to the point where the sample enters the filter. This side-stream flow shall be controlled precisely. The orientation of the flow control orifice shall allow air in the sample line to be separated from the sample before the sample enters the filter holder.
- (f) The analyser shall be equipped with a high-quality internal pressure regulator which can accommodate higher inlet pressures.
- (g) The analyser shall use standard filter media and allow fast and easy filter media replacement.
- (h) The monitors shall be housed in a suitable enclosure to protect equipment against the environment and shall be protected to NEMA 4X (IP 66).

- (i) The monitor shall employ a blinking indicator to notify the operator of the completion of the test. The SDI results shall then be read from a LCD display. The display shall be equipped with an LED backlight making it easy to see the results in any environment.
- (j) SDI tests shall be able to be activated:
 - (i) Manually from the keypad;
 - (ii) Time interval; and
 - (iii) External trigger (dry contact).

G6.12.54 Sludge Blanket Level Switches (Sludge Blanket Detector)

- (a) Sludge blanket level switches shall measure the suspended solids concentration at the sensor depth in clarifiers and sludge thickeners to determine whether the sludge blanket has reached that particular depth.
- (b) The sensor shall be corrosion resistant.
- (c) The relay unit enclosure shall be IP65 or better.
- (d) The instrument shall provide visual external indication of the status of the output via LED lights or similar.
- (e) Sludge blanket switches shall be 24V dc powered, 4-wire devices. Outputs may be PNP or relay contact.
- (f) The state of the output upon internally detected failure shall be configurable.

G6.12.55 Sludge Blanket Level Meters (Sludge Blanket Meter)

- (a) Sludge blanket level meters shall measure the suspended solids concentration at different depths to determine the sludge blanket depth in clarifiers and sludge thickeners.
- (b) The meter shall operate on the principle of sounding ultrasonically (time measuring principle) to detect suspended solids concentration and variations in transition and separation zones.
- (c) The sensor shall be permanently installed inside the sludge tank. The sensor shall use ultrasounds technology to scan the separation zones.
- (d) The sensor shall be equipped with a self cleaning device, time or operator triggered.
- (e) The meter shall comply with the following requirements:
 - (i) Submersible type;
 - (ii) Self cleaning;
 - (iii) Provide continuous reading of the separation and transition zones;
 - (iv) Shall be corrosion resistant. All the required mounting accessories (e.g. mounting pole) shall be corrosion resistant and chemically compatible with the measured media;

- (v) Sensor shall be paired with a dedicated transmitter, providing 4-20 mA HART outputs to show blanket level. There shall also be a potential free contact to indicate alarms and faults;
- (vi) Measuring range shall be 10m at least; and
- (vii) Max. measuring error: 2%.

G6.12.56 Ultra-violet Transmittance (UVT) Analysers

- (a) UV transmittance sensors shall be a continuous-reading sensor that utilise a 2-beam ultra-violet absorption measuring technology. The spectral absorption coefficient at 254 nm shall serve as the measurement of UV absorbance from which transmittance may be calculated. A reference wavelength of 550 nm shall be used for baseline compensation and turbidity correction;
- (b) The transmitter shall be enclosed in a stainless steel, cast aluminium or fiberglass reinforced plastic enclosure with mounting brackets for surface or pipe mounting. The transmitter and probe shall be furnished with all mounting hardware;
- (c) The transmitter shall be supplied with an appropriate length of signal cable for connecting the transmitter to the sensor;
- (d) Measurement range shall be 90-100% (transmittance)/cm;
- (e) Repeatability shall be: $\pm 0.5\%$ /cm; and
- (f) Accuracy shall be: $\pm 0.1\%$ /cm.

G6.12.57 Mechanical Limit Switches

- (a) Limit switches shall be heavy duty industrial mechanical units with SPDT contacts rated for 230 Vac 2A.
- (b) Cams, rocker arms, or “whiskers” type switches shall be used as appropriate for the application.
- (c) Limit switches shall be rated to IP67 or better.

G6.12.58 Proximity Switches

- (a) Proximity switches shall be of the inductive sensing type with a 3 wire 24 Vdc interface.
- (b) Suitable sensing targets shall be provided and rigidly attached in cases where the surface to be detected is non-ferrous.
- (c) Proximity switches shall have hermetically sealed cables or IP68 rated plug connectors.
- (d) Sensing distance shall normally be a minimum of 12 mm but in any event shall be commensurate with the mechanical tolerances involved.
- (e) Switches with adjustable sensitivity shall be preferred.

G6.12.59 Speed Switches

- (a) Speed switches shall be used to verify correct operation of rotation plant such as conveyors. Speed switches shall consist of:
 - (i) A proximity switch to detect the passing of a rotary or reciprocating part; and
 - (ii) A switching unit.
- (b) The switching unit shall monitor the frequency of pulses from the proximity switch and hence determine if the speed has exceeded or fallen below preset levels. Speed switch points shall be adjustable.
- (c) The switching unit shall provide a SPDT volt free contact output corresponding to each preset switch point.
- (d) The switching unit shall have indicators for power and output relay status visible with all covers closed.
- (e) Switching unit enclosure shall be rated to IP65 or better.
- (f) Speed switch sensors shall be installed so as to be able to detect as many of the possible mechanical failure modes as is practical.

G6.12.60 Warning Horns

- (a) Warning horns shall operate from 24 Vdc and shall have an enclosure rated to IP65 or better.
- (b) Warning horns shall produce a minimum sound pressure level of 95 dB at 1m but in any event shall be loud enough to be clearly audible under the likely operating circumstances.

G6.12.61 Warning Beacons

- (a) Warning Beacons shall be furnished in the specified colour, or if no colour is specified, in amber.
- (b) Beacons shall operate from 24 Vdc.
- (c) Beacons shall be 5 joule xenon strobes.
- (d) Beacons shall have an enclosure suitable for use in the process area, with a rating of IP 65.
- (e) Beacons shall have a stainless steel rod protective cage. If installed outdoors, a stainless steel sun shade shall be provided above the beacon in lieu of the protective cage.

G6.12.62 Positive Displacement Flow Totalisers

- (a) Positive displacement flow totalisers shall be of the gear type and shall have local indication only.
- (b) The totaliser shall be resettable and the enclosure shall have a rating of IP65 or better.

G6.12.63 Flammable Gas Detectors (LEL Detector)

- (a) The flammable gas detector shall continuously monitor, indicate and alarm for combustible gases.
- (b) The flammable gas detector shall be double compensated IR absorption, non-imaging optics, and 4 beam technology to detect hydrocarbons components such as methane (CH₄).
- (c) The double-compensating optical system shall be resistant towards known influences such as dust or by dirt accumulation on the optical surfaces.
- (d) IR detector and source shall provide long term stability and high resistance to vibration and shock. The optical system design shall ensure “no false alarms”, longer service intervals and a drift-free signal output.
- (e) Detector shall have RFI/EMI protection in compliance with EN 610006-2 and EN 61000-6-4.
- (f) The relationship between the hydrocarbon absorption and the detector output is inherently non-linear and therefore point flammable gas detectors shall incorporate a linearization circuit to compensate for this and produce a linear response to a broad band of gases.
- (g) The detector shall come with backlit graphical LCD display.
- (h) The detector shall have a pre-warning feature that identifies the need for maintenance while remaining full functional.
- (i) Where IR point detectors utilize mirrors they shall have design features that prevent condensation on the optical surfaces. Warning of window contamination shall be provided by a warning message on the LCD display but detectors shall remain fully functional.
- (j) The detector shall remain fully functional with a supply of power between 10V DC to 30V DC and have a start-up time of no more than 60 seconds.
- (k) The IR gas detector shall calibrated to EN60079-29-1 and EN 50271 by a Singapore recognised testing agency.
- (l) The response time shall be 1 seconds to 90% of final response (T90) following a step change in concentration.
 - (i) The detector shall interface directly to the plant PLC and shall have a 4 to 20mA output proportional to the 0 to 100% LEL calibrated to hydrocarbon gas concentration. If available, HART shall be provided.
- (m) The detector shall be suitable to be installed in hazardous explosive gas atmosphere and combustible dust environment up to Zone 1 and Zone 21 respectively, where required. The detector enclosure protection shall be at least IP67.
- (n) Where sensors are located more than 1500mm above floor level, remote type sensors shall be used, allowing the display unit to be mounted at normal accessible level.

G6.12.64 H₂S Gas Detectors

- (a) The H₂S gas detector shall continuously monitor, indicate and alarm when H₂S concentration exceeds a preset level.
- (b) The basis of measurement of Hydrogen Sulfide (H₂S) concentration shall be by electrochemical diffusion sensor.
- (c) The H₂S sensor shall be an intelligent sensor and shall automatically perform self-test functions, and provide information to allow predictive maintenance.
- (d) The sensor operating range shall be between 0 to 1000 ppm.
- (e) The sensor response time shall be T20 at <15 seconds.
- (f) The H₂S sensor shall be temperature compensated, orientation independent.
- (g) The H₂S detector shall come with graphic backlit display which shall display the measured gas concentration, selected gas type, and measuring unit during normal operation.
- (h) The H₂S detector shall be supplied with two integrated relay outputs, one for fault and one for high gas concentration.

G6.12.65 Total Solids Meters (Microwave)

- (a) The total solids meter shall continuously monitor, indicate and transmit the suspended solids content in process liquid in a pipe.
- (b) The total solids meter shall operate based on microwave phase shift technology. Measurements shall not be affected by velocity or liquid colour.
- (c) The total solids meter shall be spool-piece type, to be installed in line with process pipework.
- (d) The total solids meter measuring performance shall be:
 - (i) Range : 0 to 40% TS;
 - (ii) Sensitivity: 0.001 TS%; and
 - (iii) Repeatability: 0.01 TS%
- (e) The total solids meter shall come with field programmable electronics, self-diagnostics with troubleshooting codes and ability to program electronics with density multiplier, conductivity correction, data storage and averaging of values as a minimum.
- (f) The total solids meter shall be equipped with digital interfaces allowing for quick and easy remote interrogation of the transmitter's state.
- (g) The total solids meter shall come with backlit display which can indicate the measured suspended solids content and measuring unit during normal operation.
- (h) The total solids meter shall be factory calibrated with onsite validation of both sensor and transmitter.
- (i) The total solids meter shall a 4 to 20mA HART output.
- (j) The detector enclosure protection shall be at least IP65.

G6.12.66 Dust Analysers

- (a) The dust analyser shall continuously monitor, indicate and transmit the dust load in process gas in a pipe/duct.
- (b) Dust analysers shall be based on triboelectric effect to determine dust loads in flowing gases. The electrical charge which the dust particles experience due to friction is picked up by a probe protruding into dust channel and converted into measuring signals by electronics. The measuring signal shall be proportional to the dust concentration and calibrated at a constant gas speed.
- (c) The analyser shall be able to operate in a wide sample temperature range of up to max. 500 deg C with gas humidity <80%.
- (d) The analyser shall be able to operate in classified zone areas with relevant certificate.
- (e) The repeatability of the analyser shall be less than 2% of full scale.
- (f) Electronic span shall be done internally every 8 hours as standard.
- (g) The detection limit shall be less than 2% of full scale.
- (h) The analyser range shall be 0-100% under gas velocity of more than 5 m/s.
- (i) The analyser shall be suitable to provide 4-20mA signals for remote indication of the dust concentration on a separate control unit. The analyser shall also incorporate relays providing volt-free contacts to transmit alarms to remote panels.
- (j) The analyser enclosure shall be IP 65 or better ingress protection.

G6.12.67 Humidity and/or Temperature Meters

- (a) The humidity and/or temperature meter shall continuously monitor, indicate and transmit the relative humidity and temperature, for building automation applications.
- (b) The relative humidity sensor shall be:
 - (i) Thin film polymer capacitive;
 - (ii) Minimum accuracy of $\pm 3\%$ RH from 0 to 90% RH and $\pm 5\%$ RH from 90% to 100% RH;
 - (iii) Minimum stability of $\pm 2\%$ RH over a two year period; and
 - (iv) Transmitter shall provide output signal of 4-20mA HART, corresponding to 0 to 100% RH.
- (c) The temperature sensor shall be:
 - (i) Platinum 1000 Ω RTD;
 - (ii) Accuracy of $\pm 0.3^{\circ}\text{C}$ at 20°C ;
 - (iii) Transmitter shall provide output signal of 4-20mA, corresponding to -40° to $+60^{\circ}\text{C}$; and
 - (iv) Shall have option for calculation and output of dewpoint temperature, wet bulb temperature and enthalpy, where required.

G6.12.68 Not used in this Contract

G6.12.69 Volatile Organic Compound (VOC) Gas Analysers

- (a) VOC analysers shall be fixed photoionization detector, capable of measuring broad range of VOCs.
- (b) The analyser shall perform as per manufacturer's specification under the following operating conditions:
 - (i) Range: 0 to 100 ppm;
 - (ii) Sampling Distance: Maximum 30 meters;
 - (iii) Response Time: 40 seconds to 90% of reading using isobutylene and 40% duty cycle;
 - (iv) Output Signal: 4-20 mA output HART; and
 - (v) Alarm: Two (2) dry contacts (<30V, 2A) for low and high alarm limits.
- (c) The VOC analyser shall consists of internal diaphragm pump.
- (d) The sample gas stream shall pass through a dehumidifier prior to entering the analyser.

G6.12.70 Ultrasonic Clamp-On Flow Meters

- (a) Ultrasonic flowmeters shall be non-intrusive clamp-on type, consisting of a flow transducer and a flow display transmitter, with the following technical specifications:
 - (i) Flow Velocity Range: 0 to 12 m/s;
 - (ii) No. of Channels: 1 channel, for 1 pair of transducer inputs;
 - (iii) Accuracy: 1% of rate or better at flows > 0.3 m/s;
 - (iv) Repeatability < 0.15%;
 - (v) Current Outputs: 4 to 20mA HART output;
 - (vi) Pulse output: For totalized flow output;
 - (vii) Ingress Protection: IP65 or better;
 - (viii) Fault Indication: Fault reading shall not be indicated on the display as a zero flow, so as to differentiate a fault from a zero flow situation;
 - (ix) Display : Required;
 - (x) Diagnostics: The flowmeter shall provide self and application diagnostics to isolate any fault conditions to either equipment failure or abnormal process conditions;
 - (xi) In the event of power interruption, the equipment specified herein shall resume normal operation without manually resetting when power is restored; and

- (xii) Menu Navigation: Allows viewing of parameter settings, viewing of diagnostics, change of settings for engineering units, span range, output and logging configurations, all without disruption to flow measurement.
- (b) The flow transducers shall meet the following specifications:
 - (i) Mounting Accessories Materials: Aluminium, Stainless Steel, Textile;
 - (ii) Sensor Ingress Protection: IP68;
 - (iii) Mounting Accessories: Must be secure, and allows for easy removal for replacement of transducer couplant gel without affecting the transducer alignment and spacing;
 - (iv) Pipe Material: Transducer must be suitable for carbon steel pipe with cement lining, for raw sewage with some aeration; and
 - (v) Transducer Cables: Cables shall be non-armoured type of 15m (50 ft) length, as a continuous cable without joints.

G6.12.71 Ammonium Analysers (Colorimetric)

- (a) The analyser shall measure, indicate and transmit a signal proportional to the ammonium (NH_4^+ as N) of the process fluid.
- (b) The analyser shall employ colorimetric measuring principle.
- (c) The analyser shall include reagent and standard solutions, as well as sample preparation unit.
- (d) The measurement range shall be of 0.05 to 20, 0.5 to 50, or 1 to 100 mg/l in $\text{NH}_4\text{-N}$ in accordance with the process range.
- (e) Performance requirements:
 - (i) Accuracy: ± 2 to 3.0% of reading depending on measuring range;
 - (ii) Repeatability: $\pm 2.0\%$ of reading; and
 - (iii) Measuring batch interval: less than 5 minutes.
- (f) The analyser-transmitter shall be of the surface mounting type, protected to IP65 or better. The unit shall incorporate a liquid crystal display, a battery to maintain the cell polarising voltage in the event of power supply interruption, built-in diagnostics and simulator system with test switch.
- (g) Signal Interface: Isolated 4-20mA HART signal with two sets of voltage-free changeover contacts shall be provided for re-transmission of high and low alarm conditions.
- (h) The analyser shall be equipped with a sample preparation, i.e. sampling system as follows when specified:
 - (i) A dual membrane filtration system from the analyser manufacturer to reject solids in the samples;
 - (ii) A built-in air purge system to blow solids away from the filter membrane;
 - (iii) A reversible sample pump and air compressor shall be provided as part of the analyser. The sampling system will reverse the flow of sample and purge the sample line and filter with compressed air after each analysis cycle;

- (iv) All control components of the sampling system shall be housed and heated in an IP55-rated stainless steel enclosure, suitable for outdoor installation; and
- (v) The sampling system shall provide two programmable and potential-free contacts to detect high or low flow passing through the filter media.

G6.12.72 Carbon Monoxide Gas Analysers

- (a) Carbon monoxide gas analysers shall continuously measure the carbon monoxide content either within a process gas in a pipe/duct, or in ambient air, as shown on the Drawings. For process gas measurements, sampling systems shall comply with G6.12.75.
- (b) Analysers shall use Non-Dispersive Infra-red (NDIR) measurement to determine amount of carbon monoxide in the sample.
- (c) The instrument shall be free from interference from moisture in the gas sample.
- (d) The analyser shall require no re-calibration after initial calibration. The analyser shall perform periodic auto-zeroing to compensate for optical transmission changes.
- (e) The repeatability shall be better than +/- 1% full scale.
- (f) The accuracy shall be better than +/- 2% full scale based on the reference calibration gas certificate.
- (g) The response time shall be less than 30 sec.
- (h) The detection limit shall be 1% of full scale.
- (i) The range shall be 0-1000 ppm carbon monoxide.
- (j) The instrument shall have local indication for analogue value and alarm status and be able to provide both 4-20mA HART for remote indication of the CO concentration. The instrument shall also incorporate relays providing volt-free contacts to transmit alarms based in thresholds adjusted within the instrument.
- (k) The analyser enclosure shall be IP 65 or better ingress protection.

G6.12.73 Carbon Monoxide Gas Detectors

- (a) Carbon monoxide gas detectors shall continuously measure the carbon monoxide content either within a process gas in a pipe/duct, or in ambient air, as shown on the Drawings. For process gas measurements, sampling systems shall comply with G6.12.75.
- (b) Detectors shall use electrochemical measurement to determine amount of carbon monoxide in the sample.
- (c) The instrument shall be free from interference from moisture in the gas sample.
- (d) The detector shall require no re-calibration after initial calibration. Remote zero and span calibration functions shall be provided.
- (e) The repeatability shall be better than +/- 1% full scale.
- (f) The accuracy shall be better than +/- 2% full scale based on the reference calibration gas certificate.

- (g) The response time shall be less than 30 sec.
- (h) The detection limit shall be 1% of full scale.
- (i) The range shall be 0-1000 ppm carbon monoxide.
- (j) The instrument shall have local indication for analogue value and alarm status and be able to provide both 4-20mA HART for remote indication of the CO concentration. The instrument shall also incorporate relays providing volt-free contacts to transmit alarms based in thresholds adjusted within the instrument.
- (k) The analyser enclosure shall be IP 65 or better ingress protection.

G6.12.74 Biogas Analysers

- (a) Biogas analysers shall be used to analyse the oxygen, carbon dioxide, and methane content in the sample gas.
- (b) Electrochemical or para-magnetic analysers shall be used for the oxygen content measurement.
- (c) NDIR measurement shall be used for carbon dioxide and methane measurement.
- (d) The measurement technique shall be free from interference from moisture in the gas sample.
- (e) The analyser shall perform periodic auto-zeroing to compensate for optical transmission changes.
- (f) The repeatability of the analyser shall be better than +/- 1% of full scale.
- (g) The accuracy of the analyser shall be better than +/- 2% of full scale based on the reference gas calibration.
- (h) Calibration gas cylinder and stainless-steel pressure regulator for automatic zeroing and span adjustment shall be provided.
- (i) The response time of the analyser shall be less than 30 sec.
- (j) The detection limit shall be 1% of full scale.
- (k) The analyser range shall be 0-4 vol% oxygen, 0-20 vol% carbon dioxide and 0-75 vol% methane.
- (l) Gas sampling systems shall comply with G6.12.75.
- (m) A sample cooler shall be provided to remove moisture to avoid moisture interference for the NDIR analyser analysis.
- (n) The sample probe with remote activated 24V DC blow back valves shall be provided at the sample take-off point to overcome dirty sample gas. The blow back signal shall be provided by remote panels.
- (o) The analyser shall be able to compensate for any pressure variation in the sample gas.
- (p) The analyser shall be suitable for wall mounting and provided with local indication for analogue and alarm status and able to provide 4-20mA HART signals for remote indication of the CO₂, O₂, CH₄ concentration. The analyser shall also incorporate relays providing volt-free contacts to transmit alarms to remote panels.

- (q) The analyser enclosure shall be IP 65 or better ingress protection.

G6.12.75 Gas Sampling Systems

- (a) Where required, for gas analysers and detectors, gas sampling systems shall be provided.
- (b) Gas sampling systems shall comply with the following requirements as a minimum:
 - (i) Each stream/tag no shall have individual process isolation valve for sample inlet into sampling system;
 - (ii) A 3-way valve shall enable stream-switching between sample, zero and span gas for measurement & calibration purpose;
 - (iii) Where necessary, sample gas shall flow through cooling coil to decrease the sample temperature;
 - (iv) Where appropriate considering analyser technology, a coalescing filter shall be provided to remove excess particles & moisture and a second coalescing filter shall also be provided to remove excess moisture which might form after sample pump;
 - (v) A sample pump shall be provided to ensure sufficient flowrate of sample to analyser and reduce sample lag time;
 - (vi) A bypass line shall be provided with rotameter & regulating valve for sample lag time control;
 - (vii) A rotameter with regulating valve shall be used to adjust flow rate of sample into the analyser;
 - (viii) Sample gas conditions shall meet requirements from the gas analyser manufacturer; and
 - (ix) Used sample flows shall be discharged in a safe manner, and this may require pumping back into the main process.

G6.12.76 Pressure Gauges

- (a) Pressure gauges shall comply with BS EN 837. Pressure gauges and transmitters shall have over range protection up to 1.5 times the maximum line pressure and shall be capable of withstanding full line pressure on any side with the other side vented to atmosphere without damage or effect on the calibration.
- (b) Enclosures shall be 316 stainless steel. Internal parts shall be of stainless steel, bronze or approved corrosion-resistant material. Wetted parts shall be fully compatible with the process fluid or a suitable diaphragm seal shall be provided.
- (c) Where necessary, a diaphragm seal shall be used to segregate the gauge tube from corrosive fluid media. In ammonia applications, the diaphragm shall be in stainless steel. In chlorine applications, the diaphragm shall be in silver or tantalum. In sulphur dioxide applications, the diaphragm shall be in tantalum.
- (d) The minimum diameter for any pressure gauge shall be 150 mm unless specified otherwise or where the gauge forms part of a standard item of equipment such as instrument air filter/regulator sets.

- (e) Where compensation of more than 2% of the instrument span is needed for the difference in level between the instrument and the tapping point, the reading shall be suitably adjusted and the amount of compensation shall be marked on the dial.

G6.12.77 Acoustic Phased Array Level Transmitters (Solids Profiler)

- (a) Solids Profilers shall use an acoustic phased array for accurate level measurement of bulk solids and powders.
- (b) Solids Profilers shall have an integral array of three antennas that generate dust-penetrating low frequency acoustic waves and receive echoes from the contents.
- (c) The use of three antennas shall allow the unit to measure not only the time/distance of each echo, but also its direction.
- (d) Measurements shall be temperature compensated.
- (e) The device shall collect multiple echoes from different directions and distances to enable determination of the average level of stored material.
- (f) The acoustic frequency used shall achieve self-cleaning capabilities for the device.
- (g) Ingress protection shall be IP66.
- (h) Signal output shall be 4-20mA.
- (i) The performance of acoustic phased array level transmitters shall be as follows:
 - (i) Measurement Range : 70m;
 - (ii) Emitting Frequency : approx. 2 to 7 kHz; and
 - (iii) Beam angle : 15 to 70 degrees.

G6.12.78 Weight Transmitters (Load Cell Weighing System)

- (a) Load cell systems shall comprise a load indicator transmitter unit and load cells that operate on the strain gauge principle. The appropriate number of load cells to suit the load geometry shall be provided even though only one weight element is shown on the P&IDs.
- (b) Load cell types (e.g. compression, tension, load beam) shall be selected to suit the application. Multiple strain gauges shall be used within each load cell to provide offsetting positive and negative strain readings.
- (c) The load cell weighing transmitter shall provide a 4-20 mA dc output signal proportional to the force applied by the load. It shall be possible to utilise multiple load cells at strategically placed supporting points with a single indicator transmitter to provide weight readings for loads such as platforms, silos and tanks. The transmitter shall be able to equalize and average the readings coming from multiple load cells if required.
- (d) Load cells shall be constructed of high quality materials, such as stainless steel, suitable for the application and environment. Load cells shall be hermetically sealed to prevent the ingress of liquids and gases. Load cells shall be constructed to IP68 capable of submergence in water to a depth of 1 metre for a minimum of 4 days without affecting instrument stability or measuring performance.

- (e) Load cells shall be protected against mechanical overloads up to 250% of their rating. Electrical overloads to 150% of the load cell rating shall not cause damage or malfunction.
- (f) Indicator transmitters shall provide local indication and remote re-transmission of the following signals as described:
 - (i) Gross weight (local only);
 - (ii) Net weight (local and remote);
 - (iii) Tare weight (local only); and
 - (iv) System failure (local and remote).
- (g) If being used in a belt weighing application the system shall also provide a totalising pulse output and display with an accuracy of $\pm 0.5\%$ totalised weight or better.
- (h) The load cell measuring system shall have an overall accuracy equal to or better than $\pm 0.075\%$ of full scale.

G6.12.79 On-line Total Phosphorous Analysers

- (a) Total phosphorous (TP) analysers shall be designed for automatic continuous on-line measurement of total phosphorous content in the water sample. The analyser shall operate on the principle of UV photometric measurement at 630 nm using ascorbic acid reduction and molybdate colour solution after persulphate digestion in acidic medium.
- (b) The analyser shall be operable on 230Vac and shall comply with the following requirements:
 - (i) It shall consist of integrated on-line sampling system which allows users to calibrate, validate the analyser without removing it from the continuous sample source;
 - (ii) The measuring range shall be between 0 – 1 mg/L TP and 0-10 mg/L TP to suit the application;
 - (iii) All analyser operations shall be fully automated and shall not require user intervention. The time taken for analysis shall be up to 80 min including sample digestion of 60 min;
 - (iv) The accuracy of the analyser shall be $\pm 2\%$ of full scale range and the detection limit shall be less than or equal to 5 $\mu\text{g/L}$;
 - (v) The analyser shall have configurable automatic calibration;
 - (vi) It shall comply with APHA 4500-NO₃ (B) method; and
 - (vii) The various components of the analyser shall be housed in a wall mounted panel/enclosure rated IP65 or better.
- (c) Analysers shall have a 4-20 mA outputs. There shall also be a potential free contact to indicate alarms and faults.

G6.12.80 On-line Toxicity Analysers

- (a) Toxicity analysers shall be designed for automatic continuous on-line measurement of the relative toxicity to the biological nutrient removal process of the water sample.
- (b) The analyser shall imply a Nitrification Respiration Inhibition Test and shall be in accordance with DIN EN ISO 9509:2006.
- (c) An integrated bacteria culture shall constantly and independently produce nitrifier biomass separately from the measuring cell. The fermenter shall not be exposed to the sample stream.
- (d) For each measurement a fresh dose of biomass (bacteria) shall be used.
- (e) The Nitrification Respiration Inhibition Test measures the consumption of oxygen by the nitrifier bacteria as they convert ammonia to nitrate. Toxic substances in the sample contents inhibit the respiration of the bacteria, thus lowering their oxygen consumption. Thus a measurement of this oxygen consumption provides an indication of toxicity in the sample.
- (f) The analyser shall be operable on 230Vac.
- (g) All analyser operations shall be fully automated and shall not require user intervention. The time taken for analysis shall be up to 10 min but typically 5 minutes.
- (h) The analyser shall have a stainless steel IP65 enclosure.
- (i) Analysers shall have a 4-20 mA output and a wireless TCP/IP interface for connection to the non-process WLAN.

G6.12.81 Magnetic Level Indicators

- (a) Magnetic level indicators shall comprise a chamber, a magnetic float and an indicator mounted to the outside of the chamber. The indicator shall house a column of small flags, which flip to indicate, by colour, the level of the float in the chamber. As the float rises and lowers, the flags are flipped from one orientation to the other with the process level.
- (b) The gauge shall be suitably calibrated in percentage unless other units are specified.
- (c) The accuracy and repeatability shall each be $\pm 10\text{mm}$.
- (d) For service with injurious or clogging substances, the gauge shall be provided with flushing and drain points.
- (e) The wet materials shall be suitable for the process fluid and for the process conditions (temperature, pressure).
- (f) The exterior and housing materials shall be suitable for the environment.

END OF SECTION